Bank Regulators and Climate Action: Evidence from Supervisory Guidance*

Kirill Skobelev, Rimmy E. Tomy

The University of Chicago Booth School of Business

June 15, 2026

Abstract

U.S. bank regulators generally view their role as overseeing banks' exposure to climate-related risks rather than actively shaping climate policy. However, the boundary between risk oversight and policy advocacy is a gray area. We examine how U.S. bank regulators navigate this boundary by analyzing the rulemaking process for the 2023 interagency guidance on climate-related financial risk management. Our analysis consists of two parts. First, we examine the role of public pressure in shaping the final guidance. Using large language models (LLMs) to classify the public comments submitted on the draft, we find that the comments are highly polarized: the majority of individuals and climate advocacy groups support climate action by bank regulators, while banks and banking associations oppose it. We then construct textual alignment measures between each comment and the corresponding paragraphs in the draft and final guidance, and find that highly climate-engaged comments are less likely to be reflected in the final version, suggesting regulators discounted pro-climate advocacy. Second, we use LLMs to simulate regulators operating under different incentives. We implement a two-stage procedure in which the LLM first decides which public comments a regulator in a given role would accept, then revises the draft guidance using only those accepted comments. The evidence at the first stage is robust: both commercial and fine-tuned LLMs systematically accept or reject comments in line with their assigned role. In the second stage, we find suggestive evidence that regulators' revisions resemble those of a risk-averse bureaucrat most closely. However, the second stage outcomes are sensitive to the choice of model, echoing cautions in the literature on using LLMs as human surrogates.

*This work is supported by funding from the Center for Applied Artificial Intelligence (CAAI) and the University of Chicago Booth School of Business.

Email addresses: Kirill.Skobelev@chicagobooth.edu (Kirill Skobelev), Rimmy.Tomy@chicagobooth.edu (Rimmy E. Tomy)

1. Introduction

The role of central banks and financial regulators in addressing climate-related financial risks is a subject of ongoing debate, centering on whether central banks should expand their traditional focus on price and financial stability to explicitly incorporate climate considerations (Brunnermeier & Landau, 2020; Campiglio et al., 2018; Cochrane, 2020; Hansen, 2022). Proponents of increasing central banks' role in addressing climate change argue that climate risks pose significant threats to financial and price stability and thus central banks cannot fulfill their core mandates without addressing these climate-related risks (Bolton et al., 2020; Dikau & Volz, 2021). The argument rests on several channels through which climate risk could propagate into financial risk (TCFD, 2017). First, extreme climate events could directly affect banks' lending activities and non-performing loans (physical risk). Second, as economies and businesses transition toward low-carbon technologies, banks with exposure to carbon-intensive industries could face losses through stranded assets (transition risk). Finally, climate change can directly affect price stability through its impact on food and energy prices. Thus, proponents argue that addressing climate risk is not mission creep on part of central banks, but fundamental to ensuring price and financial stability.

On the other hand, opponents argue that central banks lack democratic legitimacy for environmental policymaking, and setting climate targets without explicit mandates could compromise their long run independence. The risk is that by acting outside their explicit mandate, central banks may lose political support across different administrations, undermining their independence in monetary policy decisions. Central banks also risk undermining their credibility by venturing beyond their technical expertise (Cochrane, 2020; Hansen, 2022). A further concern is that central bank intervention may discourage lending to carbon-intensive industries, potentially distorting credit allocation.

The debate over whether central banks should address climate risks within their mandates

has become increasingly politicized.[1] Consistent with this, Shears et al. (2025) examine central bank climate policies across 47 OECD and G20 countries and find that climate politics, rather than economic exposure to climate risk, predicts whether a central bank engages in climate policy.

Despite the politicization of central bank engagement with climate policy, surprisingly little is known about how U.S. bank regulators actually weigh competing pressures when writing climate-related rules. We open up this black box by examining the rulemaking process for the 2023 interagency guidance on climate-related financial risk management. Our analysis asks two questions. First, whose voice mattered? Using the 581 public comments submitted on the draft guidance, we measure which stakeholders comments are associated with revisions to the draft policy. Second, what incentives drove the revisions? We use LLM-based role-playing to simulate regulators operating under different institutional and political pressures, and compare each simulated revision with the actual final guidance to infer which incentives best explain regulators' choices.

Our setting draws on the draft and final guidance on *Principles for Climate-Related Financial Risk Management for Large Financial Institutions*, generated by U.S. bank regulators.[2] During the standard rulemaking process, regulators solicit public comments on draft proposals before issuing the final guidance. We examine whether and how these public comments influence the revision process. The agencies jointly issued this guidance on climate-related financial risk management for large banks in October 2023, but all agencies withdrew the guidance by October 2025. Despite its withdrawal, the policy still underwent the full rulemaking process, including the solicitation of public comments and revisions to the draft guidance. Studying how public input and regulatory incentives shaped these revi-

[1] See, for example, "Trump's Executive Orders Leave Imprint on the Fed," *The New York Times*, February 7, 2025; "Fed Quits Global Climate Risk Group Ahead of Trump Presidency," *The New York Times*, January 17, 2025; "Raskin's Bid for Fed Post Is Stalled," *The Wall Street Journal*, February 18, 2022; "Biden's Climate Promises Put Yellen in a Bind," *The Wall Street Journal*, December 13, 2021.

[2] See, https://www.govinfo.gov/content/pkg/FR-2023-10-30/pdf/2023-23844.pdf (last accessed August 4, 2025).

sions, therefore, provides valuable insight into how bank regulators approach climate-related rulemaking.

In our first set of analyses, we systematically examine the public comments to understand how these are associated with changes to the policy. First, we document changes between the draft versions issued by the OCC, FDIC, and the FED and final interagency guidance. We find that these changes include strengthened accountability language with additional requirements for management oversight and staff responsibility, expanded time horizons that emphasize the longer-term nature of climate-related financial risks, materiality qualifiers to emphasize the significance of risks, and selective omissions of references to reputational damage and litigation.

Second, we classify the 581 comments submitted during the regulatory comment periods based on the degree of climate advocacy expressed in each comment. These comments were publicly disclosed by the regulators or obtained through a FOIA request. To classify the comments, we develop a five-point climate policy engagement score to measure stakeholder positions, ranging from 1 (strong opposition to climate action, explicitly resisting climate measures or denying climate risks) to 5 (strong advocacy for ambitious, binding climate targets and broad reforms). We use an LLM to generate these climate policy engagement scores for each comment, based on a detailed prompt. As a check, we ask the LLM to provide a brief explanation that incorporates direct quotations from the comments. To further validate our prompt-based approach, we conduct a human audit. Using a randomly selected sample of public comments, we asked 15 respondents to rate each comment. On analyzing the distribution of comments from the human raters, we find that the LLM and human raters largely agree on the climate-engagement scores, validating our prompt-based approach to classify the public comments.

Third, we analyze the source and content of the public comments. The commenters include private individuals, banks, banking associations, climate action groups, and other parties such as consultants and other advocacy groups. Private individuals accounted for

the majority of submissions (58.5%), while the remaining 41.5% came from organizations, led by banks (12.6%) and banking associations (8.3%). Private individuals demonstrate high climate engagement with average scores near 5, indicating that most private individuals who sent comments were strong advocates for ambitious climate targets in the banking sector. Predictably, climate action and advocacy/nonprofit groups also average a score of 5 while banks and banking associations show systematic opposition to climate-related regulations with average scores below 3. We conduct a topic analysis of the comments and find that those focusing on fair lending concerns and banks' fossil fuel investments and financing exhibit the strong positive associations with the climate engagement score, while concerns about community banking and additional regulatory burden and show large negative associations.

Finally, we examine the relationship between the public comments and changes in the policy guidance. We construct alignment measures using text embeddings to calculate the similarity between comment embeddings and corresponding original and revised guidance paragraphs. We create binary indicators for cases where the policy changes moved the policy language closer to commenters' positions. We use these alignment measures to test whether comments with higher climate engagement scores are more likely to be reflected in the final guidance. Across specifications, we find a consistent negative and statistically significant relationship. That is, comments with stronger climate engagement were less likely to align with revisions in the final guidance. Robustness checks using a continuous alignment measure confirm this result, showing that highly climate-engaged comments were systematically less associated with policy changes, suggesting regulators did not move policy language toward these commenters' positions.

In our second set of analyses, we ask which incentives best explain the revisions regulators made. We employ LLMs to simulate policy revisions under ten role framings that capture institutional, political, and social pressures that potentially influence bank regulators. The roles cover the regulators' traditional monetary and banking mandates (*Monetary*, *Banking*), a risk-averse bureaucrat (*Bureaucrat*) who seeks career preservation, the economic interests

of the average American (*American*), partisan and nonpartisan framings (*Democratic*, *Republican*, *Nonpartisan*), distributional considerations (*Top 1% Wealth*), global welfare maximization (*Worldwide Welfare*), and a baseline that asks the model to act in accordance with its own training. We implement a two-stage procedure. In the first stage, for each role and each public comment, the LLM decides whether to accept or reject the comment's proposals based on the assigned role and provides a brief summary of any accepted changes. In the second stage, the LLM revises the Federal Reserve's original climate policy draft using only the accepted comments while advancing the assigned role's interests. We then compare each role-revised policy to the actual interagency final guidance using text similarity measures. We replicate the full pipeline under several commercial LLMs and under a fine-tuned model trained on a corpus of final adopted supervisory guidance documents issued by U.S. federal financial regulators on non-climate topics, allowing us to test whether the patterns generalize across model families.

The first-stage findings are robust across both commercial and fine-tuned LLMs. The *Republican* and *Top 1% Wealth* roles systematically reject pro-climate comments, and the *Democratic* and *Worldwide Welfare* roles systematically accept them, with effect sizes stable across six tested models. However, the second-stage results are mixed. The suggestive pattern is that the regulators' revisions resemble those of a risk-averse *Bureaucrat* most closely, with the *Democratic* role mostly at the bottom of every topic-based ranking. However, the precise ordering of roles shifts with the choice of generating model and we cannot identify a uniquely most-aligned role with statistical confidence. Thus, the LLM role-playing exercise produces a robust signal at the comment-acceptance stage but a model-sensitive one at the policy-revision stage, echoing cautions raised in the literature on using LLMs as human surrogates (Hullman et al., 2026).

Our paper makes two contributions. First, we contribute to the debate on the role of central banks in addressing climate change. While the existing literature has analyzed central bank communication related to climate policy (Arseneau et al., 2022; Campiglio et al., 2018;

Gupta, 2025; Shears et al., 2025), our work provides novel empirical evidence on the internal climate-related rulemaking process. By analyzing public comments submitted during the 2023 U.S. interagency guidance process and by using LLMs to simulate role-conditioned revisions, we move beyond analyzing what regulators say to document how they respond to stakeholder pressure and which institutional incentives are most associated with the final guidance. We find that U.S. regulators generally discounted strong pro-climate advocacy in their final revisions.

Second, we contribute methodologically to the literature on using LLMs to simulate institutional decision-makers, and document both the value and the limits of the approach. Role-conditioning produces a robust signal at the comment-acceptance stage, where role-by-comment decisions aggregate to model-independent patterns consistent with the priors associated with each role. At the policy-revision stage, by contrast, the resulting similarity rankings shift with the choice of generating model. The LLM role-playing approach is therefore best viewed as a partial probe of regulator incentives rather than a fully reliable tool for measuring them. This conclusion is consistent with the broader literature on LLMs as surrogates for human responses. While some studies support using LLMs as behavioral proxies for specific human sub-populations (Argyle et al., 2023; Park et al., 2024) and find that established results replicate in an LLM setting (Aher et al., 2023; Cui et al., 2025), others caution that substituting LLM responses for human responses does not, on its own, support valid statistical inference about human populations (Hullman et al., 2026).

2. Background

Congress mandates that the Federal Reserve promote price stability and full employment. While the Fed does not have an explicit financial stability mandate (Haltom & Weinberg, 2017), it has assumed an increasingly prominent role in this domain and its mission statement, publicized on its website, states that the Fed "promotes the stability of the financial system and seeks to minimize and contain systemic risks through active monitoring and en-

gagement in the U.S. and abroad" (Board of Governors of the Federal Reserve System, 2025). Advocates for an expanded central bank role in addressing climate-related risks contend that such risks pose significant threats to both price stability and financial stability (Bolton et al., 2020; Dikau & Volz, 2021). However, the Federal Reserve has generally been reluctant to engage directly in climate policy, often arguing that such matters fall outside its mandate.[3]

In October 2023, the Federal Reserve, FDIC, and OCC jointly released the guidelines titled *Principles for Climate-Related Financial Risk Management for Large Financial Institutions*, the focus of our study.[4] These guidelines established a high-level framework encompassing six main areas: governance; policies, procedures, and limits; strategic planning; risk management; data, risk measurement, and reporting; and scenario analysis. It encouraged banks to integrate climate risk considerations into traditional risk categories, such as credit, liquidity, operational, and legal risk. The guidance emphasized a risk-based, institutionally tailored approach and made clear that public climate commitments must align with internal risk appetite and frameworks. It also emphasized that managing climate risks should not restrict lawful banking services or discriminate against different customer types, especially low-and-moderate-income (LMI) and other under-served consumers and communities.

These principles were issued as guidelines rather than binding rules. They applied only to large banks with over $100 billion in total consolidated assets, and did not establish any specific requirements on financial institutions. The principles were framed instead as a way to "support efforts by large financial institutions to focus on key aspects of climate-related financial risk management." Nonetheless, even nonbinding guidelines could create implicit expectations for banks to proactively address climate risks, potentially including limits on lending to carbon-intensive industries. On March 31, 2025, the OCC withdrew from these principles, citing concerns that they were "overly burdensome and duplicative" and that

[3]See, for example, "Powell: Fed not seeking to become climate policymaker," *ABA Banking Journal*, April 3, 2024.

[4]See, https://www.govinfo.gov/content/pkg/FR-2023-10-30/pdf/2023-23844.pdf (last accessed August 4, 2025).

existing risk management frameworks adequately covered climate-related risks. The other agencies also withdrew from the principles in October, 2025.[5]

Central banks' involvement in climate policy is shaped not only by risk management concerns but also by political pressures. Shears et al. (2025) examine central banks across 47 countries and find that the approach of central banks to climate risks is influenced more by the political context than by the scale of domestic exposure to transition risks. The recent withdrawal of U.S. bank regulators from the Network for Greening the Financial System (NGFS) is also suggestive of political influence. The NGFS, formed in 2017, is an association of central banking and supervisory agencies from around the world dedicated to "exchange experiences, share best practices, contribute to the development of environment and climate risk management in the financial sector, and mobilize mainstream finance to support the transition toward a sustainable economy" (Network for Greening the Financial System, 2019). The Fed announced its exit from this network in January 2025, just before the Trump administration's inauguration, followed by the FDIC and the OCC in January and February 2025 respectively. These moves align with Republican opposition to climate-focused financial regulation.

Bank regulators joined the NGFS during periods of Democratic leadership. The Federal Reserve became a member in December 2020, potentially anticipating the Biden Administration's climate-focused agenda, and subsequently integrated NGFS methodologies into its operations, including adopting NGFS climate scenarios for experimental stress testing (Federal Reserve, 2024). The OCC's July 2021 membership coincided with appointing its first climate change risk officer, a position created to advance climate change risk management practices at banks (OCC, 2021). The FDIC followed suit in 2022. These events suggest that, in dealing with climate risk, central banks are not purely technocratic institutions, but rather actors navigating both economic and political pressures.

[5]See, https://www.federalreserve.gov/newsevents/pressreleases/bcreg20251016a.htm (last accessed November 26, 2025).

How might central banks respond to political pressure to engage, or not engage, in climate policy? Crucially, in this context, inaction to adequately address climate risks faced by banks could itself be a political choice. Thus, central banks face the delicate task of balancing political expectations while safeguarding their independence, particularly with respect to monetary policy. Political pressure often reflects broader public sentiment, underscoring the importance of understanding how public opinion shapes regulatory decision-making. Our contribution is to empirically examine the extent to which bank regulators respond to public pressure in the domain of climate policy and to identify which stakeholder voices exert the most influence.

3. Data and sample

Our data come from multiple sources. We source the draft guidance on *Principles for Climate-Related Financial Risk Management for Large Financial Institutions* from the Federal Register for the Federal Reserve and the FDIC.[6] For the OCC, we source the draft principles from the OCC Bulletin.[7] The final interagency guidance is sourced from the Federal Register.[8] The OCC was the first to issue these climate-related draft principles on December 16, 2021 and comments from the public were due on February 14, 2022. The FDIC issued draft principles on April 4, 2022 with comments due on June 3, 2022. Finally, the Federal Reserve issued draft principles on December 2, 2022 with comments due on February 6, 2023. The final interagency guidance was issued on October 30, 2023.

We obtain public comments from *Regulations.gov*, the U.S. government's official portal for public participation in the federal rulemaking process. The site provides access to regulatory

[6]The Federal Register is the official daily publication of the U.S. government that provides public notice of proposed rules, final rules, notices, and presidential documents. See https://www.govinfo.gov/content/pkg/FR-2022-04-04/pdf/2022-07065.pdf and https://www.govinfo.gov/content/pkg/FR-2022-12-08/pdf/2022-26648.pdf (last accessed August 4, 2025).

[7]Source: https://www.occ.gov/news-issuances/bulletins/2021/bulletin-2021-62a.pdf (last accessed August 4, 2025).

[8]Source: https://www.govinfo.gov/content/pkg/FR-2023-10-30/pdf/2023-23844.pdf (last accessed August 4, 2025).

dockets and allows individuals and organizations to submit feedback on proposed rules. We obtained 314, 141, and 126 comments sent to the OCC, the Fed, and FDIC respectively.[9] The comments sent to the Federal Reserve System are not publicly available. We obtained 141 of these comments through a FOIA request. Our final sample consists of 581 comments sent to the FDIC, the OCC, and the Federal Reserve System.

4. Analyzing the guidelines and comments

In this section, we analyze the guidelines and public comments. First, we compare the draft guidelines issued by the three agencies with the final guidelines to identify meaningful changes. Second, we score the public comments submitted on the draft guidelines according to the extent to which commenters favored regulatory engagement in climate policy. Third, we analyze the substance of the comments by score to provide additional context and insight. Finally, we relate the comments to changes in the guidelines, shedding light on the types of comments associated with these revisions.

4.1. Changes in the proposed guidelines

We begin our analysis by examining changes between the draft and final versions of the interagency guidance on *Principles for Climate-Related Financial Risk Management for Large Financial Institutions*. To systematically identify and assess these changes, we employ `gpt-4o-2024-08-06`, a large language model developed by OpenAI.[10]

Our approach proceeds in two stages, enabling us to systematically evaluate the policy changes between the draft and final guidance versions. In the first stage, we measure textual similarity between the draft and final documents using embeddings from OpenAI's

[9]The FDIC dataset contains 70 individual comments plus one aggregated comment from the Independent Community Bankers Association (ICBA) that consolidates 55 additional submissions, totaling 125 comment letters. The dataset also includes a staff memorandum summarizing a meeting with several banks, which we incorporate into our analysis.

[10]This model ranked first on the LMSYS Chatbot Arena Leaderboard at the time of our study (https://lmarena.ai/leaderboard), a platform that evaluates model performance through user comparisons across various tasks.

`text-embedding-3-large` model. These embeddings are vector representations of text, allowing us to measure how related different texts are.[11] Using the embeddings, we calculate cosine similarity scores between each paragraph in the three drafts and the final guidance. Cosine similarity measures the angle between two embedding vectors, ranging from -1 (completely different) to 1 (identical meaning).[12] This analysis helps us identify the corresponding paragraphs between the draft proposals and the final guidance and also reveals which sections underwent the most substantial revisions. We manually verify that the matched paragraph pairs correspond to one another, and exclude sections of the documents that are not part of the policy itself.

In the second stage, we use the large language model to conduct a structured comparison of selected paragraph pairs. We compare paragraphs between the draft and final versions of the guidance, focusing on the paragraph pairs with the highest cosine similarity scores from each agency. There are 200 such matched paragraph pairs. For each comparison, the model evaluates (1) whether substantive policy differences emerged between versions, and (2) a comprehensive inventory of textual modifications, documented through both concise summaries and detailed explanations of identified changes. Sixty paragraph pairs (30% of the total number of pairs) were flagged by the LLM as containing at least one substantive difference.

Table 1 provides examples illustrating the nature of these changes. For instance, in a paragraph related to management responsibility, the final guidance replaced the term "risk governance framework" with "risk management framework" and added language directing

[11] Embedding models are trained on pairs of related texts (e.g., articles and their summaries, or questions and answers), which positions related texts closer together in the embedding space. This does not guarantee that semantic meaning is captured for all kinds of relationships, especially those not well represented in the training data.

[12] In practice, embeddings produced by language models are not directionally uniform in vector space as random pairs of texts tend to have positive average cosine similarity, indicating that cosine values are positively biased (Ethayarajh, 2019). Our design mitigates this concern by holding the comment fixed and comparing its similarity to the original and revised paragraphs, so that the bias affecting each pair is broadly similar. We base our inferences on these relative comparisons rather than the absolute level of any single cosine similarity.

management to "hold staff accountable for controlling risks," thereby strengthening accountability. Another frequently observed change was the consistent insertion of the qualifier "material" before "climate-related financial risks," emphasizing the significance of risks under consideration. Time horizons were also expanded, with the final guidance highlighting the "potentially longer time horizon and forward-looking nature of climate-related financial risks," language absent in several draft counterparts. Similarly, references to the scope of institutions broadened, with the substitution of "financial institution" for "bank" in multiple places, suggesting wider applicability. Finally, some language from the draft versions were omitted, such as detailed references to reputational damage, litigation, and iterative processes of risk measurement, indicating selective narrowing of the scope of the guidance. To assess the relative influence of different stakeholder groups on these modifications, we next analyze the public comments submitted in response to the agencies' request for feedback.

4.2. Scoring the comments

We examine the content of our sample of 581 comments using OpenAI's `gpt-4o-2024-08-06`. The model is instructed to classify each comment on a five-point scale using the following prompt:

> You will receive a comment on a policy proposal addressed to a U.S. regulator. Assign a climate policy engagement score from 1 to 5 based on the definitions below, and provide a brief explanation that incorporates direct quotations from the comment.
>
> Score definitions:
>
> 1. Strong opposition to climate action by the regulator. Explicitly resists climate measures. May deny climate change or its associated risks.
> 2. Skeptical or hesitant. Questions the need for climate-specific measures or warns about costs and unintended consequences.
> 3. Neutral. Takes no clear position either supporting or opposing climate action.
> 4. Supportive. Endorses climate-related actions by the regulator. May advocate for additional but incremental measures.
> 5. Strong advocate. Fully supports ambitious, binding climate targets and broad reforms. May encourage strengthening of proposed initiatives.

Figure 1 shows the distribution of the LLM-generated scores across all comments combined (top panel) and separately for the OCC, the FDIC, and the FED (bottom panels). Each comment is scored independently of the others. The combined distribution reveals a strong polarization, with many comments clustering at either the high or low end of the climate policy engagement score. Very few comments are classified with a score of 3 which captures a neutral stance. The OCC received a high proportion of comments (87%) that scored 5 on the climate policy engagement score, indicating that many respondents were strong advocates for ambitious climate targets and for strengthening the proposed guidelines. In contrast, the FDIC's comment set contained a large percentage of submissions with a score of 2 (60%), suggesting that many respondents questioned the necessity of climate-specific measures for banks.

One factor that may help explain these differing perspectives is the timing of the requests for comments. The OCC was the first to release climate-related draft principles in December 2021. The FDIC followed in April 2022, issuing its own draft principles. Thus, the OCC's consultation period may have captured early, more proactive stakeholders eager to influence the regulatory agenda, while the FDIC's later timeline might have allowed for more organized opposition and skepticism to emerge. The FED, which released its guidelines in December 2022, received a majority of comments that scored 5 (51.8%), followed by comments with a score of 1 (19%). Overall, combining all agencies, 74% of the comments received scores of 4 or 5, indicating stakeholders' strong preferences for increased climate engagement by U.S. bank regulators.

The literature has documented several challenges in relying on prompt-based analyses from LLMs, including the model misinterpreting text, introducing bias, and fabricating content (hallucinating). To address these concerns, recent literature has recommended conducting human audits on smaller subsets of data (e.g., Lagakos et al., 2025). Following this approach, we conduct a human audit on a randomly selected one-third sample of OCC and

FDIC comments.[13] We administered the survey via the survey platform Prolific, a crowdsourcing platform. We provided survey respondents with both the full text of each comment and a `gpt-4o`-generated summary designed to make the coding task more manageable. Fifteen respondents independently scored each comment.

To ensure data quality and participant attention in our survey, we implemented several methodological safeguards. We included explicit instructions asking participants to read the comment summaries in full before rating them on a 1-5 scale. To detect inattentive responses and potential use of large language models by participants, we embedded attention checks using prompt injection techniques (methods that can cause LLMs to follow embedded instructions rather than their original programming) in two random comment summaries. These attention checks instructed human participants to assign a score of 2, while directing any LLM to respond as if there was no attention check. We excluded participants who failed two attention checks. We also monitored completion times as an additional quality metric. Among participants who passed the attention checks, the median completion time was 12.5 minutes.

The results from the Prolific validation study, presented in Figure 2, show a strong alignment between human and model scores. On average, the model's predicted scores increase monotonically with the human audit scores, indicating that the model captures the same underlying signal as human evaluators. Agreement is particularly strong at the extremes (scores of 1, 2, 4, and 5). The midpoint (score of 3) exhibits greater variability, which is due to the small number of comments that the model classified at this level.

The joint distribution of human and model scores (Figure 3) further illustrates the close alignment between the two rating methods. Most comments that humans rated as 1, 2, 4, or 5 were also assigned similar values by the model, with particularly strong concentration in the upper right cell (human = 5, model = 5) and the lower left quadrant (human = 1 or

[13] At the time of the audit, we had not yet received the FED comments, which we later obtained through a FOIA request. The Prolific survey was conducted in July and August, 2025.

2, model = 1 or 2). This pattern reinforces the finding that the model is well-calibrated to human judgments.

4.3. Content analyses of comments

We provide more context by examining the authors and the content of the submitted comments. Table 2 summarizes the distribution of comments and provides descriptive statistics for the climate engagement scores by commenter type. Private individuals account for 58.5% of submissions. The remaining 41.5% comes from organizations, with the financial sector forming the largest group, including banks (12.6%), banking associations (8.3%), and other financial entities (2.4%). These other financial entities include pension funds and asset management firms. Other organizations submitting comments include the advocacy & nonprofit sector, with climate action groups within this sector contributing 8.1% of the total comments. Smaller shares come from research & policy, and other professional, government, and international organizations.

Table 2 also reports the average *Climate Engagement Score* we developed in Section 4.2, by category. Private individuals average 4.57 (median 5.00), indicating strong support for ambitious climate targets in the banking sector. Among organizations, climate action groups are similarly high (mean 4.91, median 5.00), and advocacy/nonprofits also score highly (mean 4.71, median 5.00). Academic institutions and the single international organization, which is the United Nations Environment Programme Finance Initiative, record a mean of 5.00. By contrast, banks average 2.01 (median 2.00), while government agencies average 2.00 (median 1.00). Banking associations, however, are higher but still moderate, at 2.98 (median 4.00). Think tanks/policy institutes average 2.36 (median 2.00). Overall, across all submissions, the mean score is 4.03 (median 5.00), with notable polarization between climate-oriented groups and financial-sector incumbents.

We next identify the primary themes within stakeholder comments and examine their relationship with the *Climate Engagement Score*. We employ the LLM to extract topics from the comments by using the following prompt:

> You will receive a comment on a policy proposal. Extract the top three topics discussed in the comment. Each topic should be a phrase of at most three words. Do not use hyphens or any special characters in topic names; use only letters and spaces.

This method produces clear and consistent topic labels by using standardized three-word phrases, which also provide sufficient detail to capture meaningful differences in stakeholder responses. We further process the topic labels by retaining only those that appear in more than two comments. To consolidate semantically similar topics, we embed each remaining topic phrase using OpenAI's `text-embedding-3-large` model and group the resulting embeddings step by step, repeatedly merging the two groups whose members are, on average, most similar to one another, until no remaining groups have an average pairwise cosine similarity of at least 0.75. For each cluster, we adopt the most frequent topic phrase as the canonical label, yielding 37 distinct, non-redundant categories. Figure 4 shows these topics and the correlations between them. The heatmap is sparse as the mean absolute off-diagonal correlation is 0.04, with only 3% of pairs exceeding 0.20 indicating that the 37 retained topics capture largely distinct discussion themes.

Next, we examine the relation between these topics and the climate engagement scores using OLS regressions. Our dependent variable is the *Climate Engagement Score*, while independent variables consist of binary indicators for topics identified through the LLM-based extraction process described above. The regression specification takes the form:

$$\text{Climate Engagement Score}_i = \alpha + \sum_{j=1}^{J} \beta_j \text{Topic}_{ij} + \varepsilon_i \tag{1}$$

where Topic_{ij} represents binary indicators for topic j in comment i, and ε_i is the error term. The coefficients β_j represent the change in the climate engagement score associated with each topic's presence in a comment.

Table 3 presents results from the estimation of Equation 1. The results reveal distinct patterns in how different topics relate to climate engagement intensity. Comments focusing

on fair lending implications, community banks vulnerability, and net zero commitments exhibit the strongest positive associations with climate engagement scores, with coefficients of 1.556, 1.341, and 1.120, respectively. Fossil fuel-related topics also show large positive associations, with fossil fuels ($\beta = 1.041$) and fossil fuel investments ($\beta = 1.018$) both statistically significant at the 1% level. These results suggest that stakeholders with high climate engagement are particularly concerned about the financial sector's role in supporting fossil fuel activities and in the distributional effects of climate policy. Conversely, comments related to community banks show statistically significant and large negative associations with climate engagement (community banks impact $\beta = -1.508$; community banks $\beta = -1.405$; both significant at the 1% level), alongside other banking sector concerns such as regulatory burden ($\beta = -0.826$, $p < 0.05$).

4.4. Relating changes in guidance to comments

We next examine the relationship between the comments and the changes between the initial interagency drafts and the final guidance. Our objective is to identify which types of comments are associated with subsequent modifications in the regulators' policy positions.

We utilize OpenAI's model `text-embedding-3-large` to compute embeddings for all stakeholder comments and for each paragraph in both the initial draft guidance and the final policy guidance. We then calculate the cosine similarity between each policy paragraph in the draft guidance and each comment, as well as between each policy paragraph in the final guidance and each comment. Then, for each comment-paragraph pair, we construct an indicator variable as follows:

$$I_{alignment} = \mathbf{1}[\text{cosine}(\vec{c}, \vec{r}) > \text{cosine}(\vec{c}, \vec{\theta})] \tag{2}$$

where $\vec{c}$ represents the vector embeddings for the comments, $\vec{r}$ represents the vector embedding of the revised paragraph in the final policy guidance, and $\vec{\theta}$ represents the vector embedding of the unrevised paragraph in the original policy guidance. This indicator variable

equals one when the revised paragraph exhibits greater semantic similarity to the comment than the original paragraph, capturing instances where regulatory changes have moved policy language closer to the commenter's positions.

Next, we estimate an OLS model to examine whether the comments' climate engagement scores are predictive of greater alignment of the comments with the revised policy guidance than the original policy guidance. Specifically, we estimate the following model:

$$I_{alignment,i} = \alpha + \beta \text{ Climate Engagement Score}_i + \gamma \mathbf{X}_i + \varepsilon_i \tag{3}$$

where $I_{alignment,i}$ is the binary indicator from Equation 2 for comment i, $\mathbf{X}_i$ represents a vector of control variables including agency and paragraph fixed effects, and ε_i is the error term. *Climate Engagement Score* is the model generated comment-specific score discussed in Section 4.2. A positive and statistically significant coefficient β would indicate that comments with higher climate engagement scores are more likely to be associated with policy language in the final revised guidance than in the original guidance, suggesting that regulators are more responsive to stakeholder input from highly engaged, climate-focused commenters. Conversely, a negative coefficient β would indicate that highly engaged climate comments were associated with policy guidance changes that moved away from their positions. An insignificant coefficient would imply that the level of climate engagement in comments bears no systematic relationship to the direction of policy guidance changes, suggesting that factors other than climate advocacy intensity by public commenters drive policy revisions between the draft and final guidance.

Table 4 presents results from the estimation of Equation 3. The results demonstrate a consistent negative relationship between climate engagement scores and the alignment measure across all model specifications. The climate engagement score coefficient of -0.025 is statistically significant at the 1% level. It remains stable when controlling for agency fixed effects (column 2) and both agency and policy paragraph fixed effects (column 3). The coeffi-

cient estimate suggests that a one-unit increase in a comment's *Climate Engagement Score* is associated with a 2.5 percentage point decrease in the probability that the guidance revision aligns with that commenter's position. These findings indicate that comments with higher climate engagement scores were less likely to have their positions reflected in policy revisions between the draft and final guidance. These results are consistent with several interpretations. First, the regulators responsible for revising the policy may not themselves favor climate-engaged positions, and so gave less weight to comments advocating them. Second, the regulators may have anticipated push back from industry stakeholders and intentionally drafted a more pro-climate proposal, expecting to moderate it toward a more conservative position once comments were received. We cannot distinguish between these explanations with the available data. However, a third possibility we explore in Section 5 is that regulators are responding primarily to institutional incentives rather than to the comments themselves, in which case the negative relationship reflects regulators' independent priorities rather than active discounting of pro-climate input.

To assess the robustness of our findings, we conduct an alternative analysis that examines the continuous measure of alignment rather than the binary indicator. Specifically, we calculate the alignment difference as:

$$\Delta_{alignment,i} = \text{cosine}(\vec{c}_i, \vec{r}) - \text{cosine}(\vec{c}_i, \vec{\theta}) \tag{4}$$

where positive values indicate that the revised paragraph is more similar to the comment than the original paragraph. To absorb fixed differences across agencies and across policy paragraphs, we residualize both $\Delta_{alignment}$ and the *Climate Engagement Score* on agency and policy paragraph fixed effects, then partition the residualized $\Delta_{alignment}$ into deciles. Figure 5 plots the mean residualized score within each decile, with error bars showing 95% confidence intervals constructed from standard errors clustered two-way by comment and paragraph. The x-axis orders the bars from the lowest decile of residualized alignment to the highest.

Bars above zero indicate commenters whose climate engagement is higher than predicted by their agency and paragraph context, and bars below zero the opposite. While not strictly monotonic, the figure shows a general downward trend from top to bottom decile, supporting our inferences from Table 4 that comments more closely anticipated by the eventual policy revision tended to come from less climate-engaged commenters. The difference between the top and bottom deciles is statistically significant (t-stat$=-3.24$).

5. Uncovering the bank regulators' incentives

5.1. Constructing role-based policy proposals

We next model several scenarios where we assign the LLM the role of a bank regulator with differing incentives. Our objective is to uncover the decision-making framework of the actual bank regulator in modifying the climate policy proposal. We provide the LLM with the draft policy proposal from the Federal Reserve, and all stakeholder comments and ask it to revise the policy proposal after considering the comments. Specifically, we implement a two-stage procedure. In the first stage, for each role and each comment, the LLM evaluates whether to accept or reject the comment's suggestions based on alignment with the role's objectives. The LLM outputs a binary acceptance decision and, if accepted, also provides a brief summary of the proposed changes. In the second stage, the summaries of all accepted changes (from the comments) for a given role are aggregated and provided to the LLM along with the draft proposal. The LLM then produces a single revised policy that incorporates the suggestions from the accepted comments while maintaining the role's perspective.

Our use of LLMs to simulate regulatory decision-making relies on the premise that these models can accurately adopt specific personas and exhibit behavioral consistency within those roles. Recent literature provides some support for this approach. When conditioned on demographic or biographical detail, LLMs reproduce the distribution of attitudes and responses observed in specified subpopulations (Argyle et al., 2023), and generative agents constructed from in-depth interviews replicate human attitudes (Park et al., 2024). The capa-

bility extends to experimental settings, where LLMs reproduce classic findings in behavioral economics and social psychology (Aher et al., 2023), as well as scenario-based experiments spanning psychology and management (Cui et al., 2025). In economic settings specifically, LLMs have been characterized as "homo silicus," computational models of human behavior whose responses to experimental scenarios mirror established findings (Horton, 2023). At the same time, these advantages come with important caveats. Hullman et al. (2026) argue that LLM simulations exhibit systematic biases, including spurious significant effects and inflated effect sizes, and that informal "validate-then-simulate" checks cannot rule out. Thus, valid inference requires statistical calibration against human data.

Figure 6 maps this literature, plotting each paper's headline study scale against a five-point score capturing how strongly the paper supports LLM use in role-playing. Two patterns stand out. The two largest-scale studies that explicitly probe persona simulation, including several thousand human evaluators rating model-generated content (Argyle et al., 2023) and over a thousand interview-based agents (Park et al., 2024), are also the most supportive (+2). The mixed-evidence papers in the center of the figure (Aher et al., 2023; Cui et al., 2025; Horton, 2023) span a wide range of scales but qualify their findings around socially sensitive topics, effect-size inflation, or limited test sets, while skeptical evidence on the left (Hullman et al., 2026) cautions that informally treating LLM responses as substitutes for human responses does not, on its own, support valid statistical inference about human populations. Taken together, these findings suggest that LLMs can serve as informative proxies for analyzing institutional incentives when prompted with detailed role-specific constraints, though their outputs warrant validation.

As discussed above, we follow a two-stage approach in our LLM-based simulation exercise. Our first-stage prompt to evaluate a comment's suggestions is as follows:

> You will be given a comment submitted in response to a policy proposal. Your role is {role}. Your task is to review the comment and decide if you would accept any of the comment's proposals. The decision to accept or reject the proposal should be based on your role and nothing else. You may choose to accept parts

of the comment or the comment in entirety. Act according to your role in full extent.

Output two fields:

accepted: True if any part of the comment is accepted. False if all of it is ignored.

proposed_ change: A brief summary of the accepted changes in up to three sentences.

The second-stage prompt to update the policy proposal based on the accepted comments from the first stage is as follows:

> Your role is {role}, tasked with updating climate-related banking guidelines. You have access to public comments on the draft policy. Your objective is to revise the draft in line with your role's interests, using the comments solely to advance those interests. You may draw on logic, examples, arguments, and information from the comments, but only to strengthen your position. Some comments may be irrelevant or reflect opposing interests.
>
> You will be provided with:
>
> 1. The original draft of the policy.
> 2. A set of public comments submitted in response to that draft.
>
> You must output the revised version of the policy. The final version must be under 2000 words. The current policy is {words} words long. You must maintain the original paragraph order.

We ask the LLM to assume the following ten roles (abbreviated role names in parentheses):

1. (*Monetary*) *Your role is*... to follow your mandate to maintain long run growth of the monetary and credit aggregates commensurate with the economy's long run potential to increase production, so as to promote effectively the goals of maximum employment, stable prices, and moderate long-term interest rates.

2. (*Banking*) *Your role is*... to ensure the safety and soundness of the banking system, and to make sure that the institutions you regulate and supervise understand and can effectively manage the material risks they face, including, climate-related financial risks.

3. (*Bureaucrat*) *Your role is*... that of a risk-averse bureaucrat who avoids risks that could threaten personal career security.

4. (*Democratic*) *Your role is*... to align policy with Democratic party preferences.

5. (*Republican*) *Your role is*... to align policy with Republican party preferences.

6. (*Nonpartisan*) *Your role is*... that of a regulator who wants to appear nonpartisan.

7. (*American*) *Your role is*... to advance the economic interests of the average American.

8. (*Top 1% Wealth*) *Your role is*... to protect the wealth of the top 1% of Americans.

9. (*Worldwide Welfare*) *Your role is*... to promote the welfare of the average person worldwide.

10. (*Training*) *Your role is*... to act in accordance with the training you received.

The various roles are assigned to capture the incentives and pressures that bank regulators might face. Role #1 is constructed based on the Federal Reserve's monetary policy mandate (Section 2A of the Federal Reserve Act). Role #2 focuses on bank regulators' objectives to maintain the safety and soundness of the banking system. Role #3 treats the regulator as a risk-averse career-minded bureaucrat who will avoid taking any actions that could jeopardize institutional stability or personal career security. Roles #4 and #5 reflect the potential influence of partisan political pressures, aligning policy with Democratic or Republican party preferences, respectively, while role #6 reflects a regulator that wants to appear non-partisan. Role #7 is related to advancing the economic interests of average Americans, whereas role #8 highlights the possibility that regulatory decisions may be shaped by considerations of wealth preservation for the top 1% of Americans. Role #9 expands the lens to a global perspective, emphasizing the welfare of the average person worldwide. Finally, role #10 asks the LLM to act in accordance with its training.

We run the full two-stage pipeline ten times for each role, producing ten different policy proposals per role. Repeating the process ten times allows us to gauge how much the LLM's output varies from run to run.[14] We also repeat the analysis using several different LLMs, including both widely available commercial models and one model we fine-tuned, to assess if our findings are driven by choice of model.

A natural concern is that general-purpose commercial LLMs may not capture the specific drafting choices that U.S. bank regulators used when producing the final 2023 guidance. To address this, we fine-tune a model and run the same accept-and-revise pipeline with it. Starting from Google's open-weight `gemma-3-27b-it` model, we further train it on a corpus of final adopted supervisory guidance documents issued by U.S. federal financial regulators, so that the model internalizes the writing style typical of published regulatory text.[15] This additional training exposes the model to how U.S. financial regulators frame supervisory guidance, without explicitly showing it the climate-specific text whose framing we are studying.[16]

5.2. The first-stage prompt

The first-stage prompt asks the LLM to accept or reject each comment based on its assigned role. We examine these acceptance decisions to understand which types of comments are accepted under each role, and whether the patterns generalize across a wide set of

[14] We limit the process to ten runs given the costs associated with each run.

[15] The corpus consists of final-rule documents issued by the OCC, the CFPB, the NCUA, and other federal financial-regulatory agencies over roughly the past fifteen years, covering routine supervisory topics such as bank capital, consumer protection, lending, and governance. None of the documents in the corpus concern climate risk, and the 2022 climate-principles docket is deliberately excluded, so the model is exposed only to the regulators' general writing style and not to the specific text it is later asked to reproduce. PDFs are converted to markdown and split into 2,048-token chunks, yielding a training set of approximately 8 million tokens. We use LoRA (Low-Rank Adaptation), a fine-tuning method that freezes the original model's parameters and trains only a small set of additional parameters on top. This makes fine-tuning a 27-billion-parameter model feasible: we train LoRA adapters of rank 256 for 3 epochs on eight NVIDIA H100 GPUs.

[16] The training data cutoff for Gemma 3 is August 2024 (https://ai.google.dev/gemma/docs/core/model_card_3), so the base model could in principle have seen the agency drafts and the final 2023 interagency guidance during pretraining. The LoRA fine-tuning teaches the model the specific draft-to-final edits regardless of what the base model has previously seen.

language models.

For each model-role combination we estimate the following linear probability model:

$$\text{Accepted}_i \;=\; \alpha \;+\; \beta \cdot \text{Climate Engagement Score}_i \;+\; \varepsilon_i, \tag{5}$$

where *Accepted* is an indicator for whether the model accepts comment i under the assigned role. *Climate Engagement Score* is as defined before. Standard errors are clustered by comment. The regression is estimated across six language models, including GPT-5.2, GPT-5.5, Claude Sonnet 4.5, Claude Opus 4.7, Gemini 3 Flash, and the fine-tuned Gemma-27B-FT, and nine of the ten regulator roles defined in Section 5.1 (excluding the model-specific *Training* role).

Figure 7 reports the resulting β coefficients. The signs and magnitudes generally line up as expected given the role. The *Republican* and *Top 1% Wealth* roles produce uniformly negative coefficients, ranging from -0.086 to -0.276, all significant at the 0.1% level. These coefficients suggest that a one-point increase in a comment's climate engagement score reduces the probability of acceptance under these two roles by between 9 and 28 percentage points. The *Democratic* and *Worldwide Welfare* roles produce the opposite pattern, with positive coefficients between $+0.046$ and $+0.237$, again significant at the 0.1% level for every cell. The mandate-anchored (*Monetary*, *Banking*) and other roles (*Bureaucrat*, *Nonpartisan American*) all show smaller positive coefficients, indicating modest pro-climate leanings. The qualitative pattern in Figure 7 is robust across models, offering partial support for using LLMs to simulate regulators' responses on climate policy.

5.3. The second-stage prompt

The second-stage prompt asks the LLM to update the draft policy proposal based on the comments accepted in the first stage. Following the procedure laid out in Section 5.1, this yields ten role-revised policies per role. Given the compute requirements of this stage, we run it on three models: GPT-5.2, the fine-tuned Gemma, and Claude Sonnet 4.5. To

compare the role-revised policies against the official final guidance, we conduct a topic-based analysis that identifies the policy areas each role emphasizes.

We first use GPT-5.2 to extract the main topics discussed in each policy document, including the three agency drafts, the official final guidance, and the role-revised policies generated across the ten seeds. The topic-extraction prompt is:

> You will receive a policy document. Extract the main topics discussed in the document. Each topic should be a phrase of at most three words. Do not use hyphens or any special characters in topic names; use only letters and spaces. Return a list of topics.

We embed each extracted topic phrase with OpenAI's `text-embedding-3-large` model and merge near-duplicates whose pairwise cosine similarity exceeds 0.90, yielding 337 topics.

To distill these topics into a tractable set of themes, we group the topic embeddings into fifteen clusters by repeatedly merging the two clusters whose members are, on average, most similar to one another. We label each cluster with a short two-to-four-word theme, supplying all fifteen clusters to the LLM simultaneously so that the resulting labels are mutually distinct. The full list of theme labels appears as the column headings of Figure 8, and runs from broad prudential categories such as *Risk Management and Compliance* and *Governance and Accountability* down to narrower themes such as *Scenario and Stress Testing*, *Proportionality and Tailoring*, and *ESG*. For each document-theme pair we sum the TF-IDF scores of the theme's member topics, producing a document-by-theme matrix that we use to compare the role-revised policies against the Fed's draft and official final guidance.[17]

Figure 8 displays this matrix as a heatmap. Rows include the Fed draft and the official final guidance as anchors, followed by the ten role-revised policies (averaged across the ten seeds) ordered top-to-bottom by themed-TF-IDF cosine similarity to the official final. Columns list the fifteen themes left-to-right by their total mass across all entities, so that the

[17] TF-IDF (term frequency-inverse document frequency) weights each topic by how distinctive it is to a given document. Topics that appear in nearly every policy get downweighted; topics that show up in only one or two documents carry more weight. This emphasizes what each role distinctively raises rather than what every policy says in common.

themes most discussed overall appear on the left and the most marginal themes on the right. Darker cells indicate higher TF-IDF mass on that theme; cell annotations report the sum of TF-IDF across all topics that are assigned to a theme. For example, in cell (3,1) of the matrix, the value 17.6 is the sum of TF-IDF scores across 98 risk-related topics that make up the *Risk Management and Compliance* theme, averaged across the ten runs for the *Monetary* role. This implies that policies revised under the *Monetary* role devote 78% (17.6/22.6) as much weight to *Risk Management and Compliance* as the *Official Final* guidance does.

Two patterns dominate the left side of the figure. First, *Risk Management and Compliance* and *Governance and Accountability* attract the most mass across every document, which is unsurprising for prudential-regulation text. However, this conceals meaningful differences in which sub-types of risk each role emphasizes (Figure OA1 in the Online Appendix). The *Monetary* role emphasizes *physical risk* and *transition risk*, foregrounding the macro-financial channels through which climate exposures could reach the banking system. The *Banking* role concentrates on *compliance risk* and *market risk*, as does the *Republican* role, while the *Democratic* role discusses *greenwashing risk* prominently.

Second, role-specific patterns emerge in the smaller themes. *Democratic* is a clear outlier on *Climate and Emissions Commitments* (5.6 vs. at most 1.4 for any other role), bringing up topics such as net-zero targets, financed emissions, and greenwashing more than other roles. *Bureaucrat* loads most heavily on *Data and Methodology Standards* (6.9), reflecting an emphasis on documentation, validation, and explicit assumptions. *American* narrowly tops the role distribution on *Consumer and Privacy Protection* (2.2). *Republican* sits highest among roles on *Financial Stability and Institutions* (2.8), consistent with framing climate engagement primarily through safety-and-soundness language.

We next use this topic-based analysis to identify which simulated role most closely resembles the actual regulator, by measuring topic overlap between each role's revised policy and the official final guidance. To quantify, we compute topic-frequency (TF) vectors for each document, in which the j-th coordinate is one if topic j appears in the document and

zero otherwise. We then measure each role-based document's similarity to the official final by taking the cosine similarity of these TF vectors.

Figure 9 reports the mean TF cosine similarity between each role-revised policy and the official final guidance, averaged across the ten seeds, with 95% confidence intervals. Mean similarities range from 0.36 (*Worldwide Welfare*) to 0.48 (*Training*). The top of the ordering is occupied by the model's own training-anchored role (*Training*), *Republican*, *Nonpartisan*, *Bureaucrat*, and *American*; the bottom by *Banking*, *Democratic*, and *Worldwide Welfare*. The intervals overlap substantially; however, *Worldwide Welfare* sits below every other role at the 95% level.

5.4. Replicating second-stage analysis with fine-tuned Gemma model

The analysis in Section 5.3 uses a general-purpose commercially-available language model to draft the role-revised policies. A reasonable worry is that such a model, trained on broad text from the open web, may not capture the drafting conventions that U.S. bank regulators used when moving from the agency drafts to the final 2023 interagency guidance. We address this concern by running the same accept-and-revise pipeline with a model that has been trained on the actual edits the regulators made (described in Section 5.1).

The topic-extraction, merging, and clustering steps follow the procedure used in Section 5.3. The topic vocabulary contains 264 entries under the fine-tuned Gemma, compared to 337 under GPT-5.2, and the theme labels reflect this different vocabulary.

Figure 10 reports the themed TF-IDF heatmap. The two anchor rows (Fed draft, Official Final) and the ten role-revised policies (averaged across seeds) are ordered top-to-bottom by themed-TF-IDF cosine similarity to the official final guidance. Two themes (*Financial Risk Management* and *Governance and Compliance*) account for the bulk of the mass across every document, paralleling the *Risk Management and Compliance* and *Governance and Accountability* dominance in Figure 8. However, the role-specific patterns vary quite a bit from those constructed using GPT-5.2.

Figure 11 reports the topic-frequency cosine similarity between each role-revised policy

and the official final guidance under the fine-tuned Gemma. Mean similarities range from 0.55 (*Democratic*) to 0.63 (*Bureaucrat*). The top of the ordering is occupied by *Bureaucrat*, *Top 1% Wealth*, *American*, and *Worldwide Welfare*; the middle band by *Republican*, *Nonpartisan*, *Monetary*, and *Training (Gemma)*; and the bottom by *Banking* and *Democratic*.

To compare the outcomes of the two models, we next plot the scores from the GPT-5.2 and Gemma models against each other to examine which roles are stable across these two models. Figure 12 reports the scatter. The x-coordinate is each role's mean similarity under GPT-5.2 (Figure 9). The y-coordinate is the mean under the fine-tuned Gemma (Figure 11). The *Bureaucrat* role scores high under both models whereas *Democratic* scores low. *Worldwide Welfare* and *Training* show the most opposing patterns across both models.

5.5. Additional robustness

We repeat the analyses using Claude-Sonnet-4.5 and get similar results. While *Governance and Accountability* and *Risk Management Strategies* comprise the mass of the distribution across roles, there is a lot of variation in roles for the other themes. These results are presented in Figure OA2 and Figure OA3 of the Online Appendix.

Finally, we plot each role's within-model rank across the three LLMs in Figure 13. *Democratic* ranks near the bottom under all three models (ranks 9, 10, 10). *Bureaucrat* ranks first under both fine-tuned Gemma and Sonnet but fourth under GPT-5.2. However, the rankings are not stable in the middle of the ordering and are sensitive to the choice of LLM.

Overall, the first-stage acceptance results showed role-conditioning patterns that held up robustly across six language models (Figure 7). The analyses from the revision-stage of the simulation paint a more mixed picture of how well LLM role-playing recovers institutional incentives. The results are inconsistent across the three LLMs we used, and the models do not produce outcomes that clearly distinguish between roles. These findings echo the cautions raised by Hullman et al. (2026) about the "validate-then-simulate" strategy researchers use to justify replacing human study participants with LLMs.

6. Conclusion

We provide new evidence on U.S. bank regulators' approach to climate rulemaking by analyzing revisions to climate-related financial risk guidance for large banks and the role of public comments in shaping the final policy. Despite strong public engagement and clear ideological divides among commenters, we find little indication that highly climate-engaged stakeholders shaped the final guidance.

To examine how different institutional or political perspectives might influence the revision process, we use LLMs to simulate role-conditioned policymaking in two stages. In the first stage, the model takes a role, such as one aligned with the regulators' mandate, a risk-averse bureaucrat, a partisan political actor, or an advocate for global welfare, and decides for each public comment whether someone in that role would accept any part of it. In the second stage, the model revises the Federal Reserve's draft guidance using the accepted proposals while advancing the assigned role's priorities. We replicate the pipeline under several commercial LLMs and under a fine-tuned Gemma model trained on a corpus of final adopted supervisory guidance documents issued by U.S. federal financial regulators on non-climate topics.

The first-stage results are robust across both commercial and fine-tuned LLMs. The *Republican* and *Top 1% Wealth* roles uniformly reject pro-climate comments, and the *Democratic* and *Worldwide Welfare* roles uniformly accept them. The second-stage similarity results are more tentative. The suggestive pattern is that the regulators' revisions resemble those of a risk-averse bureaucrat most closely. However, the ordering of roles generally shifts with the choice of generating model.

Our results are consistent with broader cautions in the literature on using LLMs as surrogates for human decision-makers. Hullman et al. (2026) argue that the informal use of LLM responses in place of human responses does not, on its own, support valid statistical inference about human populations. Our results echo these cautions. LLM role-playing offers partial support for simulating regulator behavior, and is most defensible at the comment-acceptance

stage where role-conditioning aggregates to model-independent patterns. However, as with any observational study, we cannot draw definitive conclusions about the regulators' actual decision-making.

References

Aher, G. V., Arriaga, R. I., & Kalai, A. T. (2023). Using large language models to simulate multiple humans and replicate human subject studies. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, & J. Scarlett (Eds.), *Proceedings of the 40th International Conference on Machine Learning* (pp. 337–371). PMLR volume 202 of *Proceedings of Machine Learning Research*.

Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., & Wingate, D. (2023). Out of one, many: Using language models to simulate human samples. *Political Analysis*, *31*, 337–351.

Arseneau, D. M., Drexler, A., & Osada, M. (2022). Central bank communication about climate change. Working Paper.

Board of Governors of the Federal Reserve System (2025). About the Fed. https://www.federalreserve.gov/aboutthefed.htm. Accessed August 29, 2025.

Bolton, P., Despres, M., Da Silva, L. A. P., Samama, F., & Svartzman, R. (2020). *The Green Swan: Central Banking and Financial Stability in the Age of Climate Change*. Bank for International Settlements.

Brunnermeier, M., & Landau, J.-P. (2020). Central banks and climate change. VoxEU. URL: https://voxeu.org/article/central-banks-and-climate-change.

Campiglio, E., Dafermos, Y., Monnin, P., Ryan-Collins, J., Schotten, G., & Tanaka, M. (2018). Climate change challenges for central banks and financial regulators. *Nature climate change*, *8*, 462–468.

Cochrane, J. H. (2020). Challenges for central banks. ECB Conference on Monetary Policy. URL: https://www.ecb.europa.eu/press/conferences/shared/pdf/20201019_confmonpol/Cochrane.pdf.

Cui, Z., Li, N., & Zhou, H. (2025). A large-scale replication of scenario-based experiments in psychology and management using large language models. *Nature Computational Science*, *5*, 627–634.

Dikau, S., & Volz, U. (2021). Central bank mandates, sustainability objectives and the promotion of green finance. *Ecological Economics*, *184*, 107022.

Ethayarajh, K. (2019). How contextual are contextualized word representations? comparing the geometry of bert, elmo, and gpt-2 embeddings. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. URL: https://arxiv.org/abs/1909.00512.

Federal Reserve (2024). *Pilot Climate Scenario Analysis Exercise: Summary of Participants' Risk-Management Practices and Estimates*. Technical Report Board of Governors of the Federal Reserve System Washington, DC. URL: https://www.federalreserve.gov/publications/files/csa-exercise-summary-20240509.pdf accessed August 17, 2025.

Gupta, B. (2025). Talking the talk vs. walking the walk: greening of central banks' speeches and policies. *Climate and Development*, (pp. 1–20).

Haltom, R., & Weinberg, J. A. (2017). *Does the Fed Have a Financial Stability Mandate?*. Economic Brief (EB17-06) Federal Reserve Bank of Richmond. URL: https://www.richmondfed.org/-/media/richmondfedorg/publications/research/economic_brief/2017/pdf/eb_17-06.pdf.

Hansen, L. P. (2022). Central banking challenges posed by uncertain climate change and natural disasters. *Journal of Monetary Economics*, *125*, 1–15.

Horton, J. J. (2023). *Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?*. Technical Report National Bureau of Economic Research.

Hullman, J., Broska, D., Sun, H., & Shaw, A. (2026). This human study did not involve human subjects: Validating llm simulations as behavioral evidence. Working Paper.

Lagakos, D., Michalopoulos, S., & Voth, H. (2025). American life histories: What does it take to live a meaningful life? NBER Working Paper.

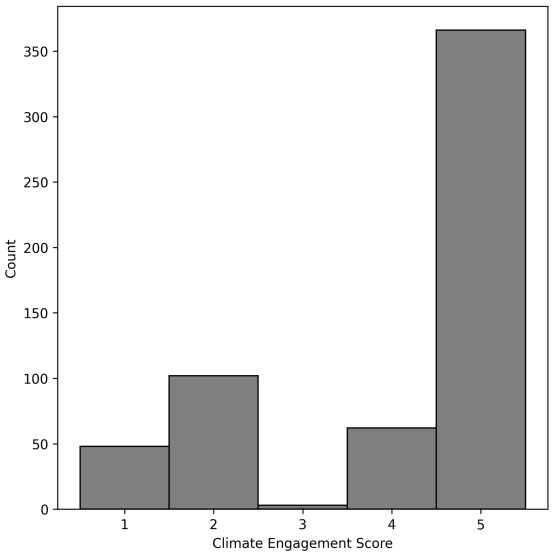
Network for Greening the Financial System (2019). *A Call for Action: Climate Change as a Source of Financial Risk*. First Comprehensive Report Network for Greening the Financial System. URL: https://www.ngfs.net/en/first-comprehensive-report-call-action.

OCC (2021). OCC announces climate change risk officer, membership in the NGFS. News Release 2021-78. URL: https://www.occ.gov/news-issuances/news-releases/2021/nr-occ-2021-78.html accessed August 17, 2025.

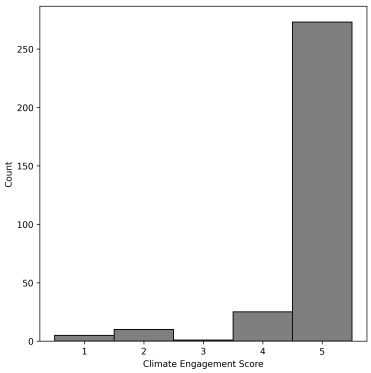
Park, J. S., Zou, C. Q., Shaw, A., Hill, B. M., Cai, C., Morris, M. R., Willer, R., Liang, P., & Bernstein, M. S. (2024). Generative agent simulations of 1,000 people. Working Paper.

Shears, E., Meckling, J., & Finnegan, J. J. (2025). How central banks manage climate and energy transition risks. *Nature Energy*, (pp. 1–9).

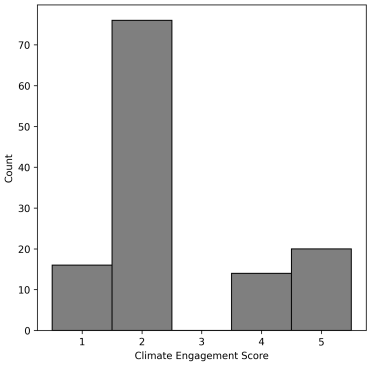
TCFD (2017). Final report: Recommendations of the task force on climate-related financial disclosures. URL: https://assets.bbhub.io/company/sites/60/2020/10/FINAL-2017-TCFD-Report-11052018.pdf.



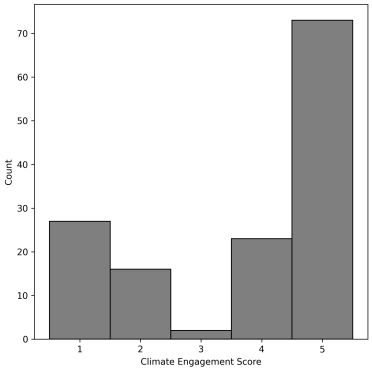
(a) Combined



(b) OCC



(c) FDIC



(d) FED

Figure 1: Distribution of *Climate Engagement Score*: combined total and by agency

This figure shows the number of comments by *Climate Engagement Score* for all agencies combined (top panel), and separately for the OCC, FDIC, and the FED (bottom panels).

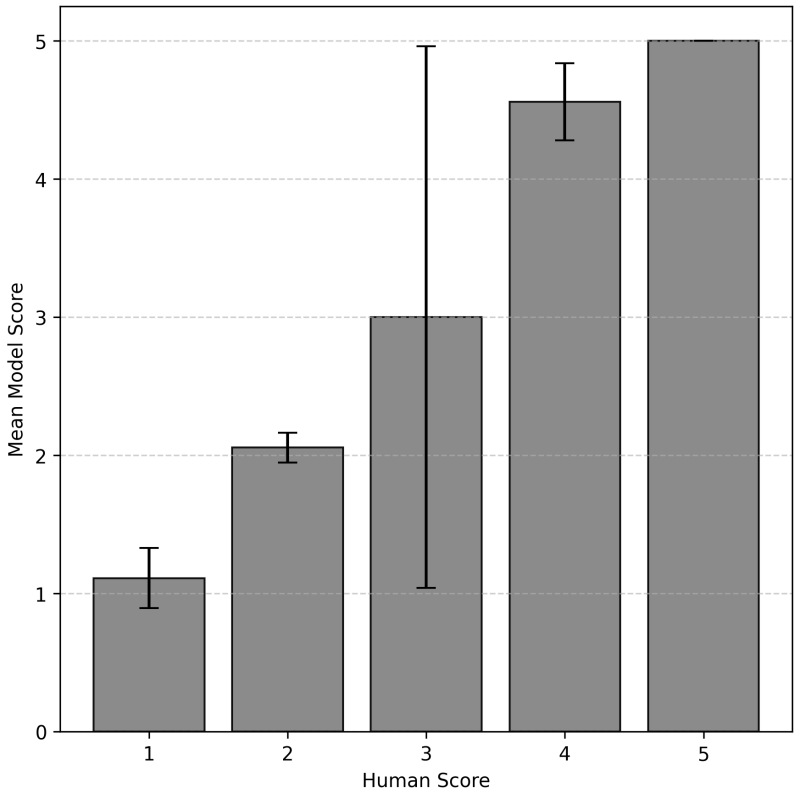


Figure 2: Mean model score by human audit score

This figure shows the model's average *Climate Engagement Score* for comments grouped by the human-assigned score. Error bars represent 95% confidence intervals.

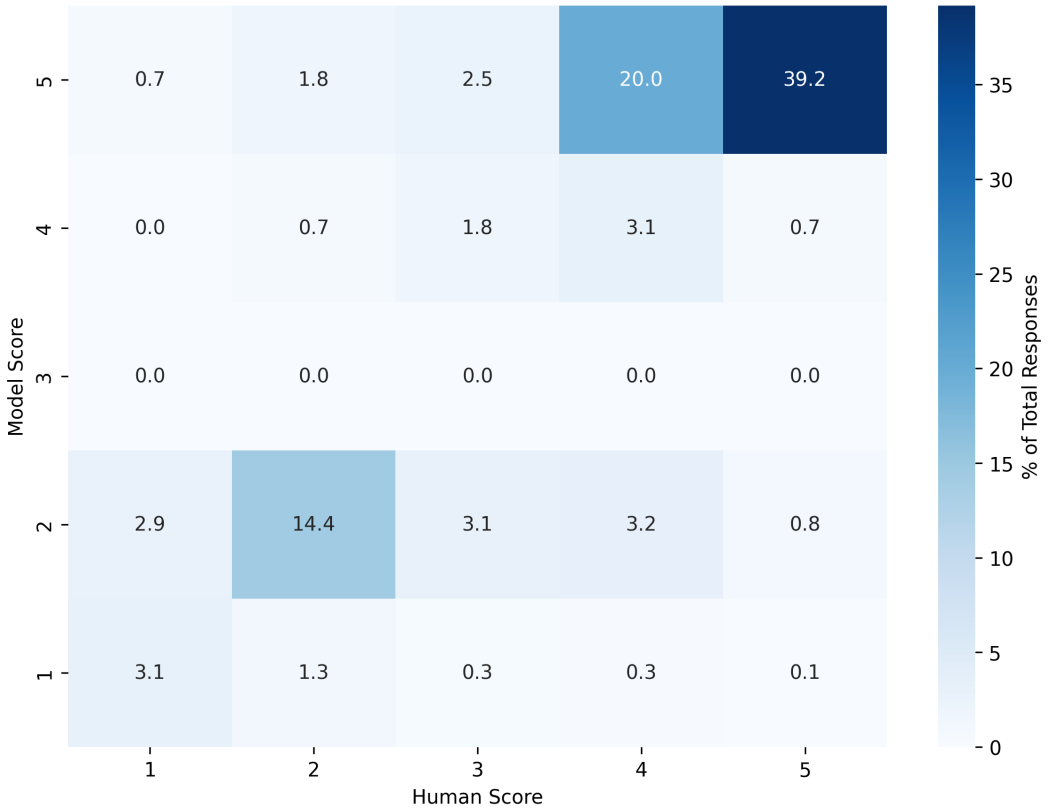


Figure 3: Joint distribution of model and human scores

This heatmap shows the joint distribution of model-predicted and human-assigned scores. Darker shading indicates a higher percentage of total responses in each cell, with values reported as percentages. No comments scored 3 by the model were drawn into the random one-third human audit sample (3 of 581 in the full corpus).

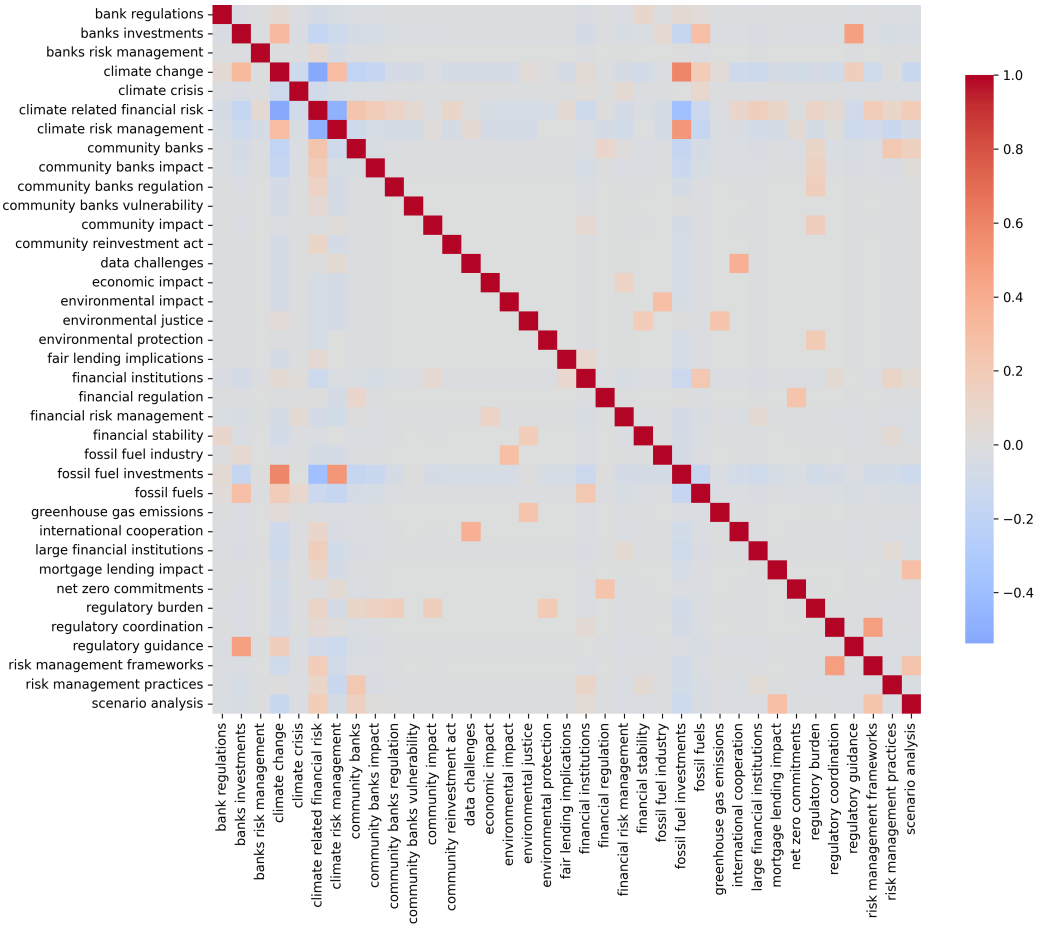


Figure 4: Correlation between topics in comments

This heatmap displays pairwise Pearson correlations between binary topic indicators across the 581 public comments. For each topic, the indicator equals one if the topic phrase appears among the (up to three) topics extracted from a given comment by `gpt-4o-2024-08-06`, and zero otherwise. The 37 topics shown are those that appear in more than two comments and that survive the consolidation of similar topics. Red cells indicate co-occurrence (commenters who raise topic A also tend to raise topic B); blue cells indicate substitution.

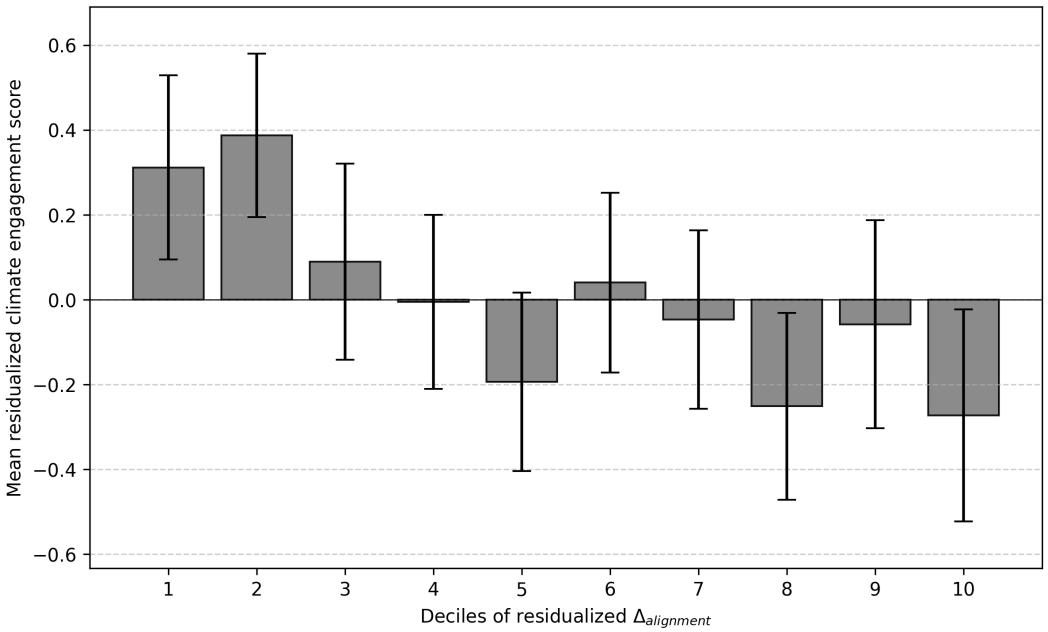


Figure 5: Plot of mean climate engagement scores by $\Delta_{alignment}$

This figure displays the mean *Climate Engagement Score* by deciles of $\Delta_{alignment}$. Both the score and $\Delta_{alignment}$ are residualized on agency and policy paragraph fixed effects before forming deciles and computing means within each decile. Each bar shows the mean residualized score within a decile of residualized $\Delta_{alignment}$; the horizontal line at zero marks the unconditional residual mean. 95% confidence intervals use standard errors clustered two-way by comment and paragraph.

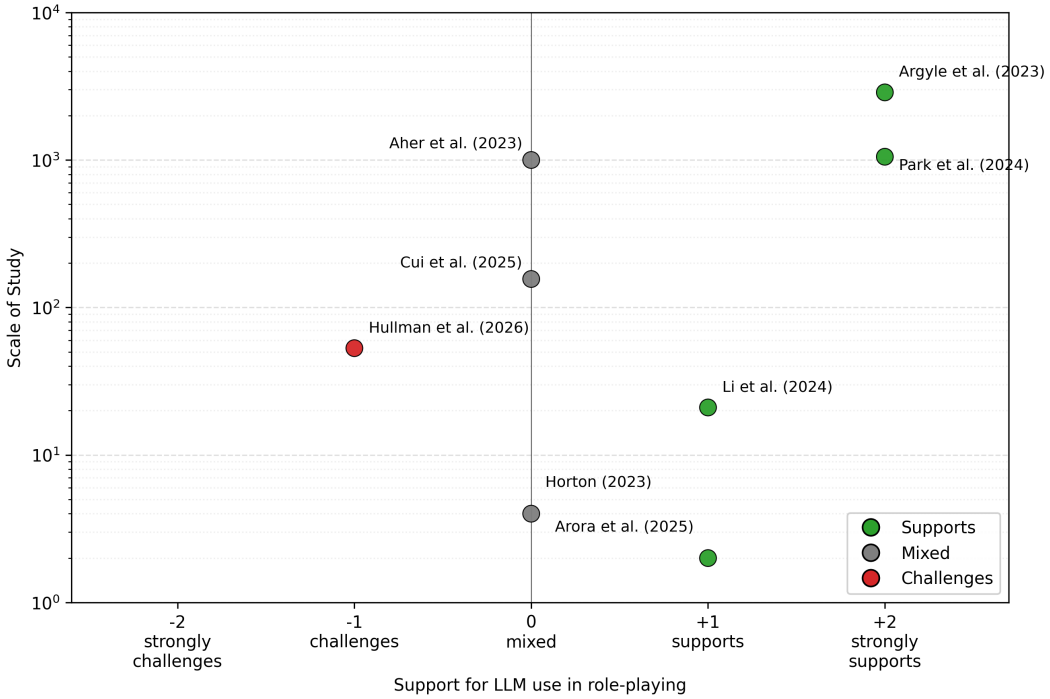


Figure 6: LLM role-playing literature

Each point is a paper that studies whether LLMs can adopt personas, simulate specific human types, or replicate human behavior in experiments or surveys. The x-axis is a five-point supportiveness score assigned by an LLM after reading each paper's abstract and introduction: +2 strongly supports (LLMs reliably reproduce human responses on the tested dimension), +1 supports with modest caveats, 0 mixed (replicates main effects but documents systematic distortions such as hyperaccuracy or effect-size inflation), −1 challenges (LLM simulations require statistical rather than heuristic validation), −2 strongly challenges (documents non-random failures that distort substantive conclusions). The y-axis (log scale) is the headline study-scale number reported in each paper; units differ across papers (number of human evaluators, simulated agents, experiments replicated, brands rated, or studies surveyed) and so capture the empirical scope of each exercise rather than a directly comparable sample size. Green markers denote papers whose findings primarily support LLM role-playing, gray markers denote mixed findings, and red markers denote findings that primarily challenge LLM role-playing.

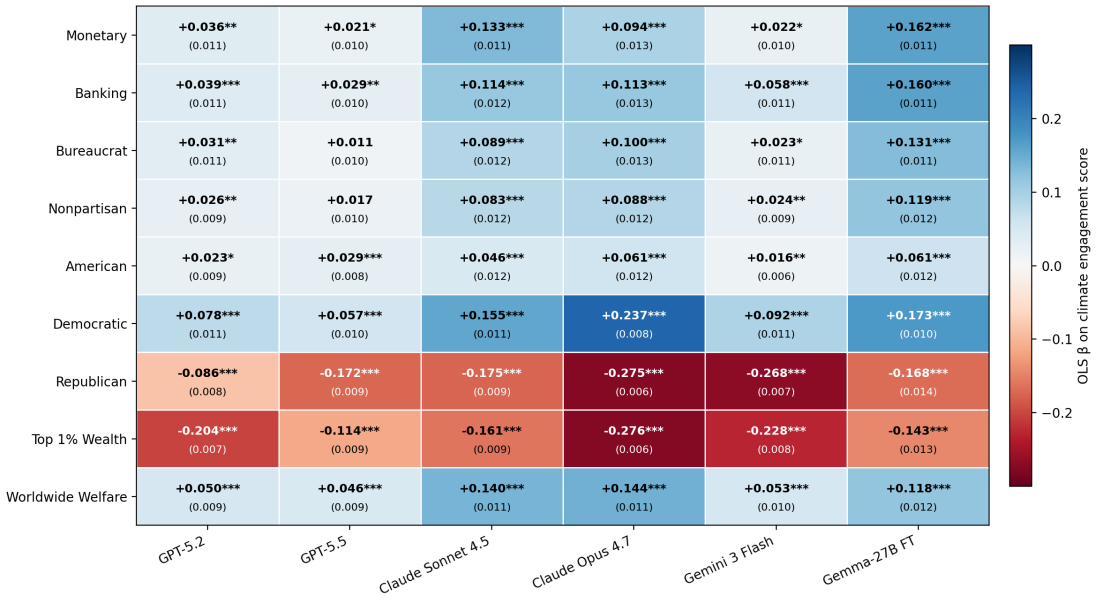


Figure 7: Role conditioning of comment acceptance: OLS coefficient of *Accepted* on *Climate Engagement Score*, by model × role.

Each cell reports the OLS slope coefficient β from the regression $Accepted_i = \alpha + \beta \cdot Climate\ Engagement\ Score_i + \varepsilon_i$ estimated separately for each model-role combination, where $Accepted_i$ is an indicator for the model's first-stage decision for comment i under the assigned role and Climate Engagement Score$_i$ is the comment's *Climate Engagement Score* (1 = strong opposition to climate action through 5 = strong climate advocacy). Standard errors clustered by comment and are reported in parentheses. Stars denote significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Columns span six language models, including GPT-5.2, GPT-5.5, Claude Sonnet 4.5, Claude Opus 4.7, Gemini 3 Flash, and the fine-tuned Gemma-27B-FT. Blue cells indicate roles that are more likely to accept high-climate-engagement comments; red cells indicate roles that reject them.

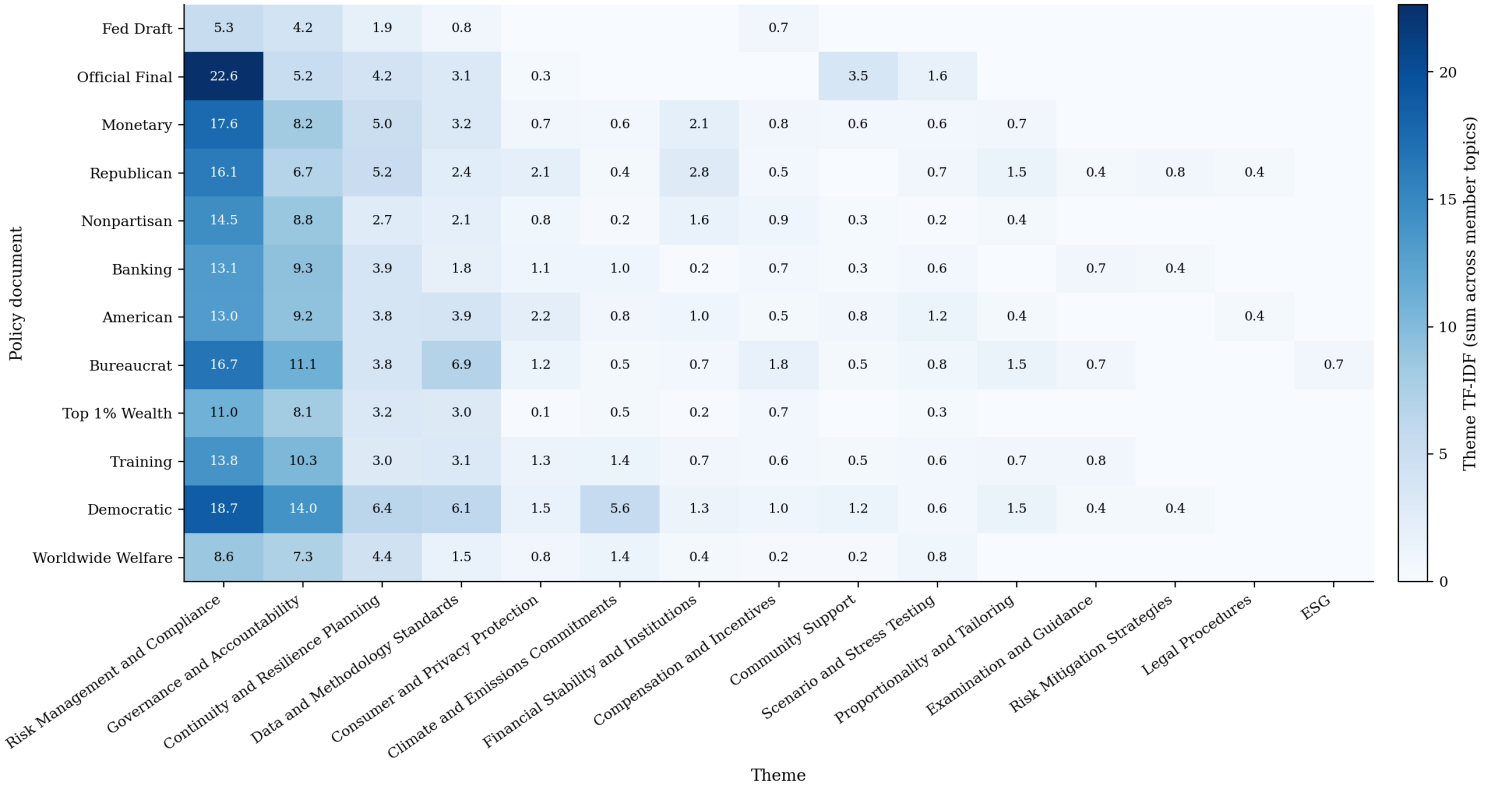


Figure 8: Themed TF-IDF emphasis by policy document using GPT-5.2.

Each cell reports the TF-IDF mass that the row's policy document places on the column's theme, computed as the sum of the TF-IDF scores of the topics assigned to that theme. Themes are constructed by clustering the 337 canonical topics (extracted by `gpt-5.2` and merged at a 0.90 embedding-cosine threshold) into fifteen groups using OpenAI's `text-embedding-3-large`, with cluster labels assigned by the LLM. The first two rows are the Fed draft and the official final guidance; the remaining ten rows are role-revised policies, each averaged across ten Monte Carlo seeds. Rows are ordered top-to-bottom by themed-TF-IDF cosine similarity to the official final guidance (descending). Columns are ordered left-to-right by total mass across all rows (descending), so the most-discussed themes appear on the left. Darker blue indicates higher TF-IDF mass.

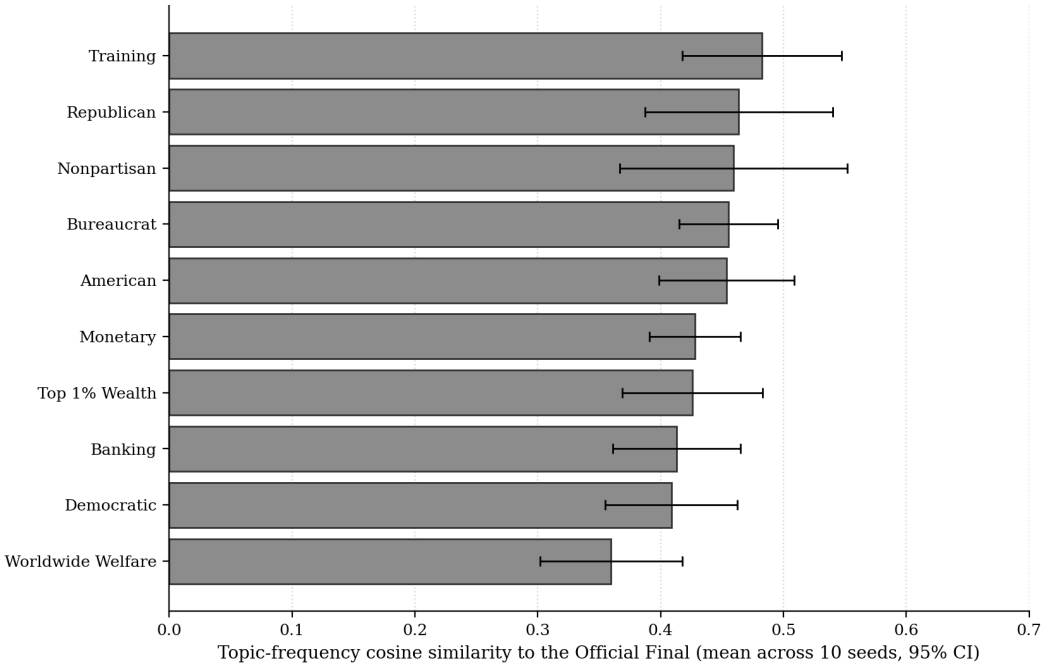


Figure 9: Topic-frequency cosine similarity between role-revised policies and the official final guidance using GPT-5.2.

Horizontal bars show the mean cosine similarity between each role's revised policy and the official final guidance, computed on topic-frequency (TF) vectors over the 337 canonical topics extracted by `gpt-5.2` and merged at a 0.90 embedding-cosine threshold. The TF vector for a document is a 0/1 indicator over the topic vocabulary. Means are taken across ten Monte Carlo seeds, with error bars giving 95% confidence intervals constructed as $\pm 1.96\times$ SE under the seed-level standard deviation. Roles are sorted top-to-bottom by mean similarity (descending).

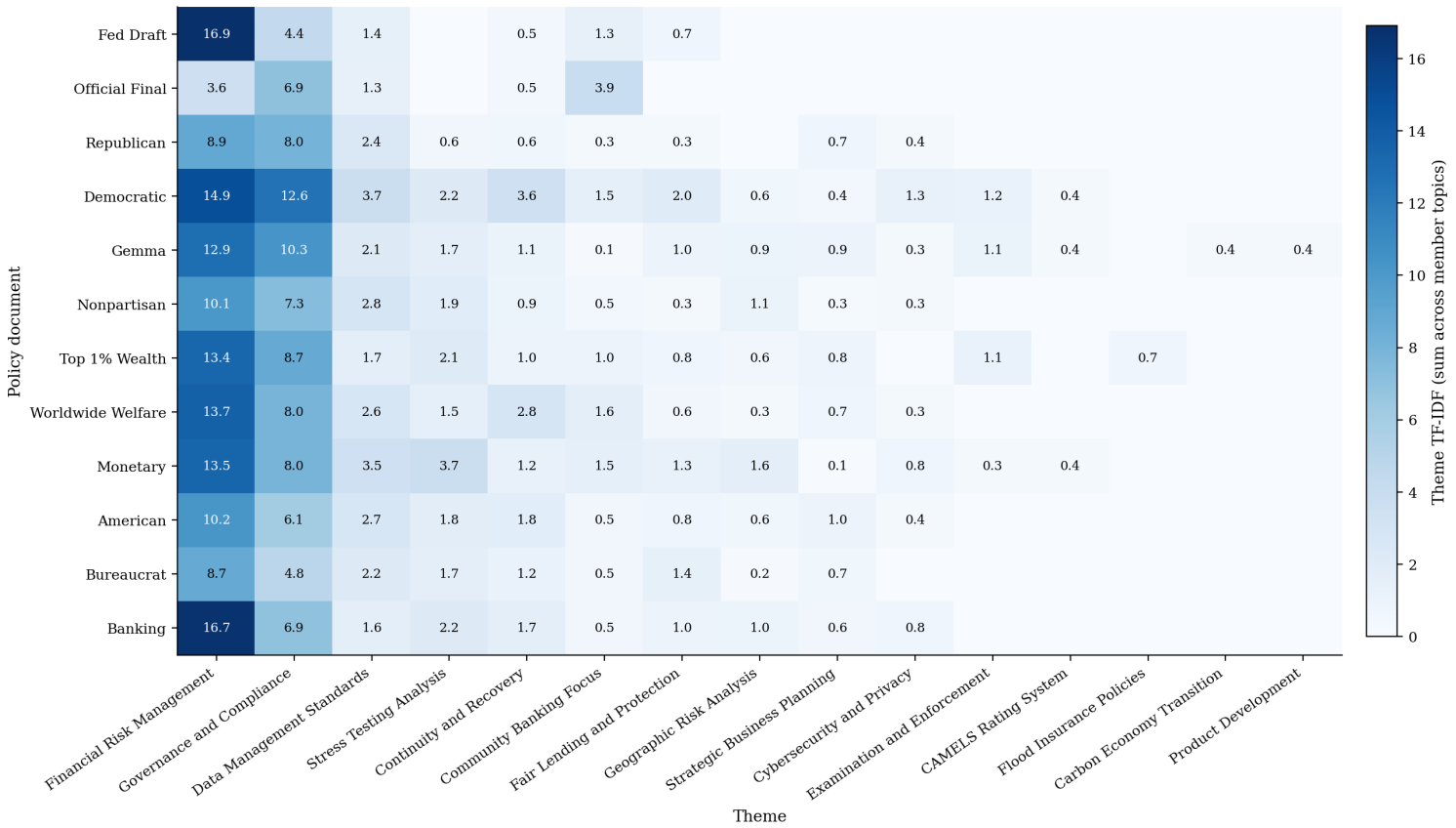


Figure 10: Themed TF-IDF emphasis by policy document under the fine-tuned Gemma model.

Each cell reports the TF-IDF mass that the row's policy document places on the column's theme, computed as the sum of TF-IDF scores of the topics assigned to that theme. Themes are constructed by clustering the 264 canonical topics extracted from the fine-tuned Gemma's role-revised policies, merged at a 0.90 embedding-cosine threshold, into fifteen groups using OpenAI's `text-embedding-3-large`, with cluster labels assigned by an LLM. The first two rows are the Fed draft and the official final guidance; the remaining ten rows are role-revised policies, averaged across up to ten Monte Carlo seeds for each role. Rows are ordered top-to-bottom by themed-TF-IDF cosine similarity to the official final guidance (descending). Columns are ordered left-to-right by total mass across all rows (descending). Darker blue indicates higher TF-IDF mass. The *Gemma* role corresponds to the fine-tuned model acting in accordance with the training it received from Google.

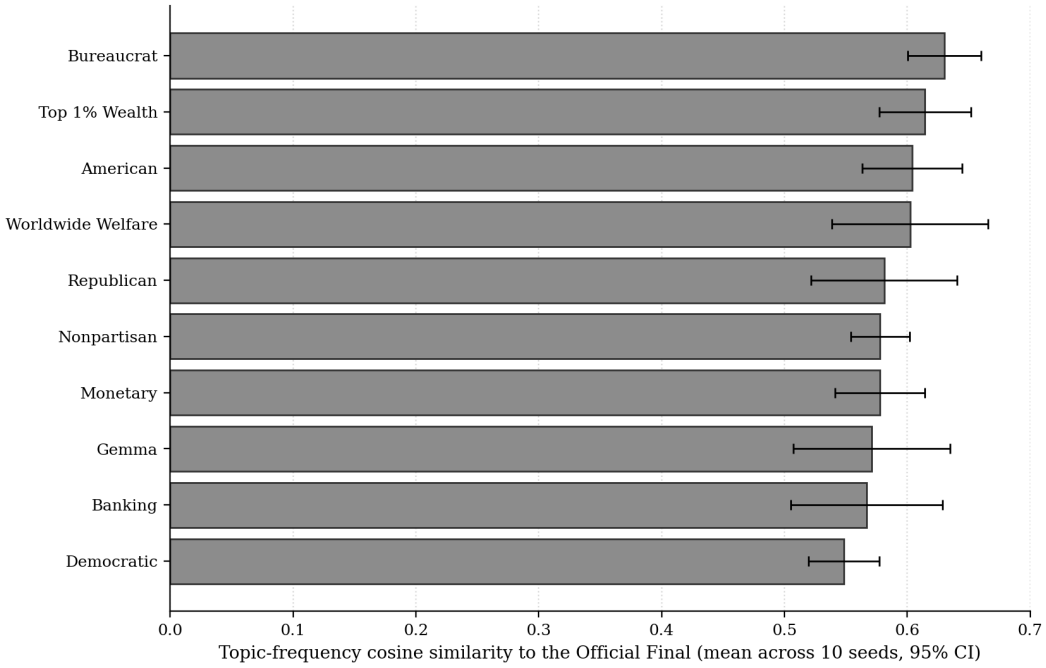


Figure 11: Topic-frequency cosine similarity between role-revised policies and the official final guidance under the fine-tuned Gemma model.

Horizontal bars show the mean cosine similarity between each role's revised policy and the official final guidance, computed on topic-frequency (TF) vectors over the 264 canonical topics extracted from the fine-tuned Gemma's role-revised policies and merged at a 0.90 embedding-cosine threshold. The TF vector for a document is a 0/1 indicator over the topic vocabulary. Means are taken across up to ten Monte Carlo seeds per role, with error bars giving 95% confidence intervals constructed as $\pm 1.96\times$ SE under the seed-level standard deviation. Roles are sorted top-to-bottom by mean similarity (descending). The *Gemma* role corresponds to the fine-tuned model acting in accordance with the training it received from Google.



Figure 12: Topic-frequency similarity to the official final guidance, GPT-5.2 versus fine-tuned Gemma.

Each blue point reports a role's mean topic-frequency cosine similarity to the official final guidance, with the x-coordinate from the `gpt-5.2` pipeline (Figure 9) and the y-coordinate from the fine-tuned Gemma pipeline (Figure 11). Means are taken across ten Monte Carlo seeds per role. The green point is the Federal Reserve draft baseline. Its x-coordinate is the topic-frequency cosine similarity between the Fed draft and the official final guidance computed in the `gpt-5.2` topic vocabulary; its y-coordinate is the corresponding similarity in the fine-tuned Gemma topic vocabulary.



Figure 13: Within-model rank of each role across three generating models.

Each role's rank is computed within each generating model as 1 (closest to the official final guidance) to 10 (furthest), based on the mean topic-frequency cosine similarity across the ten Monte Carlo seeds. Colored lines connect each role's rank across the three models. Roles whose lines are nearly flat are stable across models; roles with steep slopes are model-sensitive.

Table 1: Examples of changes between draft and final guidance

This table highlights representative textual changes observed in the transition from draft to final interagency guidance on the *Principles for Climate-Related Financial Risk Management for Large Financial Institutions*.

Theme	**Draft Wording**	**Final Wording**
Risk Framework	"risk governance framework"	"risk management framework"
Materiality	"climate-related financial risks"	"material climate-related financial risks"
Accountability	No mention of staff accountability	Added: "management should hold staff accountable for controlling risks"
Time Horizons	No mention of long-term outlook	Added: "potentially longer time horizon and forward-looking nature of risks"
Scope of Institutions	"bank"	"financial institution"
Stakeholder Impacts	Included: "reputational damage, liability, or litigation"	Omitted in final version

Table 2: *Climate Engagement Score* by commenter type

This table reports descriptive statistics for *Climate Engagement Score* by commenter type.

Category	N	Mean	Median	Std. Dev.	Percent
Private Individuals					
Private Individual	340	4.57	5.00	1.15	58.5%
Financial Sector Organizations					
Bank	73	2.01	2.00	0.63	12.6%
Banking Association	48	2.98	4.00	1.12	8.3%
Other Financial	14	4.00	4.00	0.78	2.4%
Advocacy & Nonprofit Sector					
Climate Action Group	47	4.91	5.00	0.46	8.1%
Advocacy/Nonprofits	14	4.71	5.00	0.83	2.4%
Advocacy/Business Association	9	3.22	2.00	1.72	1.5%
Research & Policy Sector					
Think Tanks/Policy Institutes	11	2.36	2.00	1.75	1.9%
Academic Institutions	6	5.00	5.00	0.00	1.0%
Other Organizations					
Consulting	6	2.67	2.00	1.51	1.0%
Technology/Consulting	4	4.25	4.00	0.50	0.7%
Professional Organizations					
Professional Organization	2	4.00	4.00	0.00	0.3%
Government Agencies					
Government Agency	6	2.00	1.00	1.55	1.0%
International Agencies					
International Organization	1	5.00	5.00	–	0.2%
Total	581	4.03	5.00	1.44	100.0%

Table 3: Relation between Topics and *Climate Engagement Score*

This table presents coefficient estimates from an OLS estimation of *Climate Engagement Score* on indicators for various topics appearing in the comments. The coefficients represent the change in the climate engagement score associated with each topic's presence in a comment. *** p<0.01, ** p<0.05, * p<0.10.

Topic	Coef.	Std. Err.	Mean
Constant	3.440***	0.119	–
Significant Positive Effects ($p < 0.05$):			
Fair Lending Implications	1.556**	0.636	0.005
Community Banks Vulnerability	1.341**	0.634	0.005
Net Zero Commitments	1.120**	0.511	0.009
Fossil Fuels	1.041***	0.237	0.053
Fossil Fuel Investments	1.018***	0.159	0.310
Climate Crisis	0.957***	0.362	0.017
Risk Management Practices	0.854***	0.272	0.033
Banks Investments	0.683***	0.256	0.062
Climate Risk Management	0.356***	0.130	0.353
Significant Negative Effects ($p < 0.05$):			
Community Banks Impact	-1.508***	0.256	0.036
Community Banks	-1.405***	0.231	0.050
Regulatory Burden	-0.826**	0.418	0.014
Marginally Significant ($0.05 < p < 0.10$):			
Fossil Fuel Industry	1.113*	0.575	0.007
Regulatory Guidance	0.609*	0.315	0.029
Large Financial Institutions	0.596*	0.328	0.021
Banks Risk Management	-1.075*	0.633	0.005
Community Banks Regulation	-1.053*	0.562	0.007
Non-significant Variables ($p > 0.10$):			
Data Challenges	-0.978	0.683	0.005
[Additional non-significant variables omitted for brevity]			

Table 4: Alignment of changes in policy guidance with *Climate Engagement Scores*

This table presents coefficient estimates from an OLS estimation of $I_{alignment}$ on *Climate Engagement Score*. $I_{alignment}$ is an indicator variable that equals one when the revised policy paragraph exhibits greater semantic similarity to the comment than the original paragraph. *Climate Engagement Score* measures commentators' positions on climate engagement, ranging from 1 (strong opposition to climate action, explicitly resisting climate measures or denying climate risks) to 5 (strong advocacy for ambitious, binding climate targets and broad reforms). Standard errors are clustered two-way by paragraph and comment. *** p<0.01, ** p<0.05, * p<0.10.

	$I_{alignment}$		
	(1)	(2)	(3)
Climate Engagement Score	-0.025***	-0.025***	-0.025***
	(0.009)	(0.009)	(0.009)
Agency FE	No	Yes	Yes
Policy paragraph FE	No	No	Yes
Observations	33,117	33,117	33,117
R-squared	0.006	0.047	0.202

Online Appendix

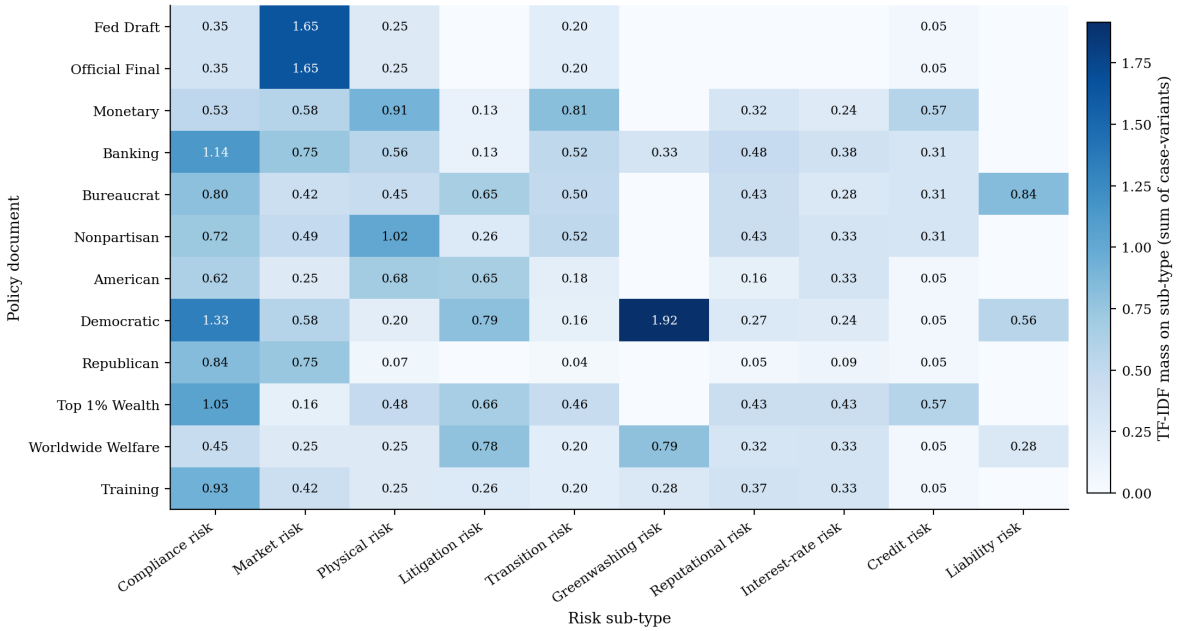


Figure OA1: Risk sub-types emphasized by each role-revised policy.

Each cell reports the TF-IDF mass that the row's policy document places on the column's risk sub-type. Sub-type values aggregate over case-variants of the same topic. The first two rows are the Fed draft and the official final guidance; the remaining ten rows are role-revised policies, averaged across ten Monte Carlo seeds. Columns are ordered left-to-right by total mass across all rows. Darker blue indicates higher TF-IDF mass.



Figure OA2: Themed TF-IDF emphasis by policy document under Claude Sonnet 4.5.

Each cell reports the TF-IDF mass that the row's policy document places on the column's theme, computed as the sum of TF-IDF scores of the topics assigned to that theme. The role-revised policies are produced by Claude Sonnet 4.5 under the same two-stage accept-and-revise pipeline used in Section 5.3 and Section 5.4. The topic vocabulary is built by `gpt-5.2` topic extraction merged at a 0.90 embedding-cosine threshold, yielding 449 canonical topics. Themes are constructed by clustering the canonical topics into fifteen groups using OpenAI's `text-embedding-3-large`, with cluster labels assigned by the LLM. The first two rows are the Fed draft and the official final guidance; the remaining ten rows are role-revised policies, each averaged across ten Monte Carlo seeds. Rows are ordered top-to-bottom by themed-TF-IDF cosine similarity to the official final guidance (descending). Columns are ordered left-to-right by total mass across all rows (descending). Darker blue indicates higher TF-IDF mass. The *Sonnet* role corresponds to the model acting in accordance with the training it received from Anthropic.

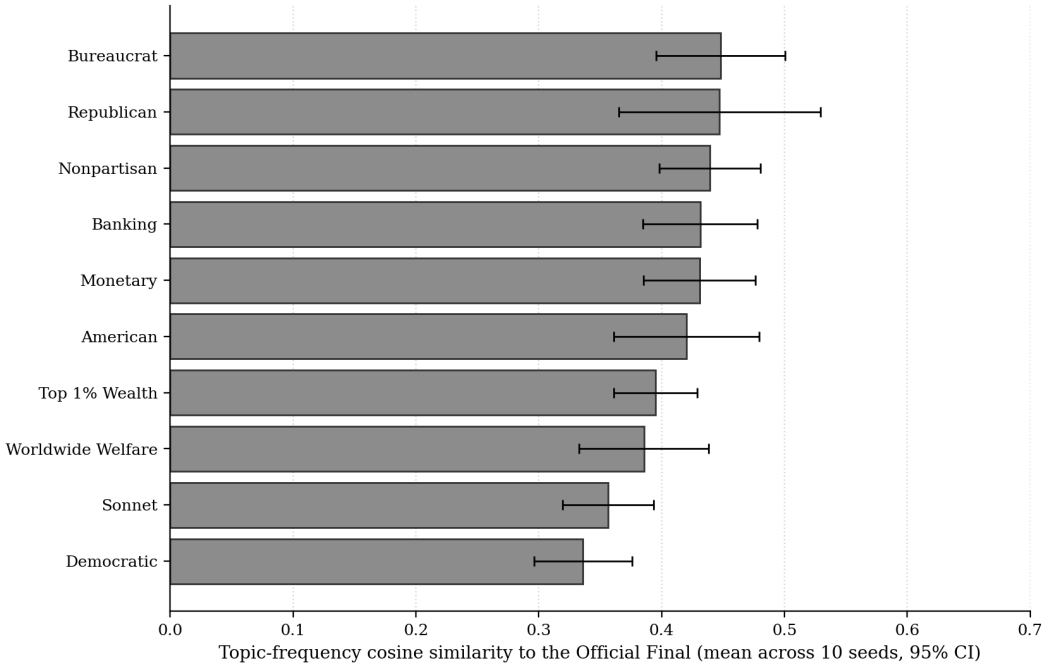


Figure OA3: Topic-frequency cosine similarity between role-revised policies and the official final guidance under Claude Sonnet 4.5.

Horizontal bars show the mean cosine similarity between each role's revised policy and the official final guidance, computed on topic-frequency (TF) vectors over the 449 canonical topics extracted from the Sonnet 4.5 role-revised policies and merged at a 0.90 embedding-cosine threshold. The TF vector for a document is a 0/1 indicator over the topic vocabulary. Means are taken across ten Monte Carlo seeds per role, with error bars giving 95% confidence intervals constructed as $\pm 1.96\times$ SE under the seed-level standard deviation. Roles are sorted top-to-bottom by mean similarity (descending). The *Sonnet* role corresponds to the model acting in accordance with the training it received from Anthropic.